When Innovation Outpaces Policy: Hyperscale Data Centers and the Local Costs of Investment Incentives*

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Abstract

Governments increasingly offer tax incentives to attract capital-intensive projects with the goal of generating tax revenue and creating jobs. While these policies attract investment, whether they generate local returns and how these returns vary with project size remains an open question. Exploiting the staggered adoption of state data center tax incentives and county-level variation in proximity to major water bodies, I study how these incentives affect public finances, employment, local resources, and political outcomes. I report three key findings. The increase in spending includes services and amenities for local residents, particularly on schools. Second, employment rises in complementary industries, but overall employment falls. This pattern, accompanied by increases in local resource prices, is consistent with data centers competing for scarce resources and crowding out employment in other industries. Third, the local economic and political effects differ before and after the transition toward larger data centers. States that adopt incentives early experience limited effects on electricity rates and farmland prices, and local incumbents are more likely to be reelected. In states that adopt incentives following the transition to larger facilities, exposure raises electricity rates and developed land prices, lowers farmland prices, and makes incumbents less likely to be reelected. These findings show that incentives for capital-intensive investment can strengthen local public finances while generating unintended economic and political consequences as subsidized projects grow larger.

Keywords: Data Centers; Public Finance; Employment; Local Politics; Energy Demand; Land Price; Regional Economic Activity

JEL Codes: G38; H25; H71; H72; R3; R11; R38; O33; Q41

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1 Introduction

The rapid expansion of capital-intensive technologies and recent policy proposals have renewed debate over using tax incentives to encourage capital investment.[1] These incentives have become both increasingly prominent (Acemoglu et al., 2020) and contentious.[2] Capital investment incentives condition tax benefits primarily on the purchase and ownership of capital, rather than on specific local outcomes such as job creation (Ku et al., 2020; Neumark and Grijalva, 2017), worker training (Abramovsky et al., 2011; Martins, 2021), or environmental remediation (Swenson, 2019). Therefore, tax relief can increase with the scale of a project even when employment and other local benefits do not, creating unintended consequences. Despite their widespread adoption, the extent to which these policies benefit local economies remains an open question.

This paper examines these effects by asking two broad questions about capital investment policy. First, how do large capital-intensive projects affect local economies, governments, and residents? Second, how do these consequences change as technological innovation increases the scale and resource intensity of subsidized investment? These questions are central to evaluating modern place-based policies and incentives that target rapidly evolving capital-intensive industries.

Policymakers justify these incentives as a means of increasing local employment and expanding the tax base,[3] but whether they achieve either objective is not obvious ex ante. First, their benefits depend on whether they induce investment that would not otherwise occur (Hall and Jorgenson, 1967; House and Shapiro, 2008; Zwick and Mahon, 2017; Maffini et al., 2019; Ohrn, 2019). Second, a larger tax base may strengthen local fiscal capacity, increase spending on amenities for residents, and lower municipal borrowing costs by improving perceived credit quality (Schwert, 2017; Gao et al., 2020). These fiscal benefits, however, are not guaranteed. Tax exemptions reduce tax revenue from new capital, and subsidized projects may require additional spending on infrastructure and public services.

Governments may finance these costs by issuing debt, which could strain debt capacity and raise borrowing costs (Chava et al., 2024). Alternatively, they may increase taxes or cut spending on local amenities (Bartik, 2019).

Third, even when these incentives increase investment, their employment effects depend on whether new capital creates complementary jobs or displaces existing workers (Garrett et al., 2020). Finally, subsidized projects may lead to agglomeration (Dougal et al., 2015) and broader local multiplier effects (Moretti, 2010). At the same time, they may compete with incumbent firms and residents for scarce resources, raising resource prices and displacing existing economic activity. The consequences for host communities are therefore uncertain even when these incentives succeed in attracting additional investment.

Technological change complicates these predictions because investment incentives may remain unchanged as the scale and resource requirements of the projects they attract evolve. Larger projects can generate greater tax revenue and complementary economic activity, but these benefits may not increase proportionately with investment. Employment may grow more slowly than capital, while demand for electricity, water, land, and public infrastructure may increase substantially. Therefore, local effects of an investment incentive change in magnitude or direction as projects become larger and more resource-intensive. This creates a unique challenge when incentives are designed for emerging industries. Incentives designed for projects in their nascent stages may remain in place for successive generations of projects that are substantially larger, replace equipment more frequently, and consume more local resources than originally anticipated, generating unintended local consequences as that industry develops.

[1]A 2025 U.S. executive order directed the Secretary of Commerce to develop a program providing financial support including loans, grants, and tax incentives for data centers and related infrastructure.

[2]Examples of statutory investment incentives include accelerated depreciation under Internal Revenue Code (I.R.C.) § 168(k) and elective expensing of qualifying depreciable property under I.R.C. § 179. Sections 13201 and 13101 of the Tax Cuts and Jobs Act of 2017 expanded these provisions, respectively. The CHIPS Act of 2022 created an investment tax credit under I.R.C. § 48D for qualified property used in semiconductor manufacturing facilities.

[3]The sales- and use-tax exemption passed in Washington in 2022 stated that the policy was intended to encourage investment, thereby "growing employment in the technology industry while adding real and personal property to state and local property tax rolls, thereby increasing the county tax base." 2022 Wash. Sess. Laws ch. 267, §§ 1–2; see also Wash. Rev. Code §§ 82.08.986–82.08.9861 and 82.12.986–82.12.9861.

I study these questions through the expansion of data centers in the United States, using the industry as a lens on capital-intensive development more broadly. Data centers require upfront and recurring capital expenditures, and policymakers often see them as opportunities to expand the local tax base, create jobs, and position their regions within the growing digital economy. Many states have adopted targeted incentives that reduce the cost of data center investment through sales and property tax exemptions for equipment, facilities, electricity, and other qualifying inputs (Table B.2) to attract these investments. As of 2025, 40 states offer some form of data center incentive (Figure 1). This makes data centers one of the most heavily subsidized categories of modern infrastructure.

There are several challenges associated with evaluating investment tax incentives. First, many investment policies are adopted nationally, leaving little geographic variation with which to construct a comparison group. Second, these incentives are often enacted as part of broader legislative packages, making it difficult to isolate their effects from those of other policy changes introduced at the same time.[4] To address these challenges, I exploit the staggered adoption of data center incentives across states. For each adoption year, I construct a separate cohort sample containing states that adopt in that year and states that have not yet adopted or never adopt an incentive. I then combine these cohorts in a stacked design (Gormley and Matsa, 2011).

This approach, however, leaves open the possibility that states adopting these policies differ from other states or have concurrent policy changes. To mitigate this concern, I use within-state variation in counties' exposure to the policy, comparing water-adjacent and non-water-adjacent counties under the same state-level policy environment. I classify counties that contain or border a coast, lake, or river as water-adjacent counties. I use water-adjacent status as a predetermined suitability measure because access to water is important for cooling large data centers.

[4]For example, the Tax Cuts and Jobs Act of 2017 expanded investment expensing alongside reductions in corporate income tax rates, new limits on interest deductions, and changes to the taxation of foreign earnings. Similarly, the CHIPS and Science Act of 2022 introduced a semiconductor investment tax credit alongside direct manufacturing subsidies, funding for research and development, and workforce training.

The specification includes three sets of fixed effects. County-by-cohort fixed effects account for time-invariant county characteristics, such as physical geography and proximity to major population centers. I also add state-by-year-by-cohort fixed effects to account for contemporaneous statewide changes and water-adjacent-by-year-by-cohort fixed effects to account for changes that affect water-adjacent counties more broadly, such as changes in federal environmental regulations, flood-insurance policy, or water-adjacent land values. After accounting for these factors, the estimate compares the change in outcomes between water-adjacent versus non-water-adjacent counties in adopting states with the corresponding change in control states. The identifying parallel trends assumption is that, absent the adoption of data center tax incentives, the relative outcomes of water-adjacent versus non-water-adjacent counties would have evolved similarly across treated and control states.

I begin by examining whether the empirical design captures increases in data center development activity. I use novel data center facility data from Aterio, supplemented with hand-collected information from Baxtel.com, on new projects and facility details, including announcement dates, estimated space, and power capacity. The estimated increase in data center count equals 30.3 percent of the mean used to normalize the outcome. The corresponding increases in facility space and power capacity equal 42.3 and 50.2 percent of their respective normalization means. These results suggest that incentives attract more data centers and increase total data center capacity.[5]

To answer the paper's first question of how large, capital-intensive projects affect local economies, governments, and residents, I examine two margins: local public finances and employment.

Following incentive adoption, water-adjacent counties in adopting states experience a 2.0 percent increase in total revenue relative to the comparison counties. This increase comes primarily from a 2.5 percent increase in total taxes and a 4.4 percent increase in revenue from government services, although the latter is not significant. I then examine

[5]One concern is that the subsidies may also benefit Bitcoin mining facilities, which resemble data centers and may use water for cooling. I find no increase in the number of Bitcoin mining facilities after incentive adoption, consistent with policies that distinguish data centers from cryptocurrency mining operations.

which tax categories contribute to this increase. Sales tax receipts rise by 5.5 percent, and property tax receipts increase by 1.9 percent, while public utility taxes and license taxes do not change significantly. These findings suggest that despite receiving exemptions on qualifying data center purchases and property, incentives can expand components of the local tax base through non-exempt purchases or induced economic activity. I also analyze the sources of the increase in revenue from government services. I find that utility revenue increases by 4.7 percent. This increase in utility receipts is consistent with greater demand for publicly provided utilities as the water-adjacent counties attract capital-intensive projects.

Interestingly, revenue gains are accompanied by higher expenditure. Total expenditure rises by approximately 3.1 percent, but the composition of this spending is particularly informative. I find a significant increase in spending on services and amenities for residents. Spending on services—including education, parks, public safety, and health—increases by 2.3 percent. Among these services, education spending rises by 1.8 percent, with school construction spending increasing by 8.2 percent. I also find a modest improvement in math scores, although reading scores do not change significantly. Together, these findings suggest that the fiscal gains are accompanied by investments that ultimately benefit residents.

Total long-term debt does not change significantly following incentive adoption. However, examining the changes in its composition reveals an interesting finding. I examine both private-purpose debt, which governments issue to finance private-sector activities such as industrial development, private hospitals, and housing, and debt for governmental purposes, which supports public activities such as schools, roads, and utilities. Debt for governmental purposes increases by approximately 5.5 percent, while the estimate for private-purpose debt is negative but statistically insignificant. This suggests a reorientation of borrowing toward public infrastructure and services rather than a broad expansion of indebtedness. Municipal bond yields and spreads nevertheless rise by about 11 and 10 basis points, respectively, suggesting that fiscal gains coexist with higher borrowing costs, potentially reflecting uncertainty about future revenues and ongoing expenditure

commitments.

The labor market results highlight an important trade-off. Water-adjacent counties experience a 1 percent decline in private employment, a 0.75 percent decline in establishments, and a 0.53 percent decline in average wages relative to control counties. However, these aggregate declines mask gains in sectors directly connected to data center construction and operation. Employment increases in power generation and natural-gas distribution. Similarly, HVAC and Semiconductor industries experience a 1 percent increase in employment. At the same time, employment in residential construction declines by 3.1 percent. I assign industries as either complementary, resource-intensive or other and analyze how the effects vary. I find that employment, establishments, and wages increase in complementary industries while decreasing in resource-intensive industries. This is consistent with data center development crowding out industries that compete for electricity, water, and land. These findings suggest that data center development reallocates employment across industries rather than generating broad-based employment growth.

Finally, I examine how data center development affects local resource prices. Data centers may increase residential electricity prices by raising demand and requiring additional grid investment, but regulated rates and negotiated power contracts may limit the costs passed on to households. Data center development may also increase developed-land prices by creating demand for construction-ready sites. Its effect on agricultural land is less clear: farmland may become more valuable because of its potential conversion to industrial use, or less valuable if data center development raises concerns about water availability. I find that data center exposure increases developed-land prices by approximately 19.8 percent, consistent with greater demand for development-ready property. In contrast, the average effects on residential electricity and agricultural-land prices are small and statistically insignificant.

To answer my second question—whether the local consequences of data center investment vary with project scale—I examine heterogeneity across subsidy adoption cohorts. In

2016, the expansion of cloud computing increased the need for larger data center facilities to store data and provide computing services to businesses (Shehabi et al., 2016). I find evidence consistent with this change in my sample: median facility size increased from approximately 58,000 square feet before 2016 to 250,000 square feet in 2016 or later, while median power capacity rose from 5 MW to 66 MW. I therefore examine this heterogeneity by comparing observations before 2016 with those in 2016 or later, when data centers became substantially larger and more resource-intensive.

Before 2016, exposure to data center incentives is associated with little change in residential electricity prices or agricultural land values, and no detectable decline in employment. After 2016, however, the same policy exposure is associated with higher residential electricity prices, a sharp decline in agricultural land values, and lower local employment. Developed land values increase in both periods, consistent with stronger demand for sites suitable for data center development. I also examine how data center development affects local elections. Incumbents are more likely to be reelected in states that adopted subsidies before 2016, but less likely in states that adopted them in 2016 or later.

To mitigate concerns that adopting and non-adopting states differ systematically in economic characteristics or trends, I assess their comparability in two ways. First, I compare distributions of observable economic characteristics and find that the two groups are broadly similar (Figure A.1). Second, I compare state-level economic trends over the sample period and find similar patterns (Figure A.2).

Another concern is that water-adjacent counties may differ systematically from non-water-adjacent counties in characteristics that independently affect local outcomes. I compare demographic and economic characteristics in 2000, the first year of treatment, and find no statistically significant differences across various observable characteristics(Table B.3).

Incentive adoption may also reflect lobbying by firms planning to locate there, raising concerns about the timing of adoption. These location decisions may respond to economic conditions that also affect my outcomes. I identify states where firms lobbied for data

center incentives using hand-collected evidence from news reports and local government meeting records. My main results remain robust when I exclude these states.

I also address concerns that data center incentives are adopted as part of broader incentive packages. Other provisions in these packages may affect local outcomes independently of data center investment. I examine the legislation and timing of each subsidy adoption to identify incentives enacted as part of a larger package and find similar results after dropping these states.

Local governments may offer their own subsidies which coincide with state incentive adoption and disproportionately benefit water-adjacent counties, confounding my estimates. I use differences in local authority under Home Rule and Dillon's Rule to address this because identifying individual local incentives is difficult, as disclosure is often incomplete and negotiations frequently involve non-disclosure agreements. Home Rule gives local governments discretion over local affairs, while Dillon's Rule limits their authority to powers granted by the state. I restrict the sample to states whose counties operate under Dillon's Rule, where local discretion is more constrained, and incentives depend on state authorization. The positive effects on data center investment and fiscal outcomes persist.

Next, I examine whether major data center hubs or highly populated counties drive the results by excluding Virginia and California and, separately, the 10 and 20 most populated counties. I also test sensitivity to the choice of event window. My results are robust to these restrictions. Finally, I address concerns about concurrent changes in water-related activity in water-adjacent counties in adopting states. I run placebo tests on outcomes related to water access but not directly targeted by data center incentives and find no clear changes around adoption (Figure A.3).

This paper contributes to the literature on place-based policies and business tax incentives.[6] One strand of this literature studies discretionary, firm-specific subsidies offered by state and local governments to attract individual firms or projects (Bartik, 1994; Greenstone and Moretti, 2003; Greenstone et al., 2010; Kline and Moretti, 2014; Bartik, 2019;

[6]Slattery and Zidar (2020) provide an in-depth analysis of different place-based policies.

Chava et al., 2024). This literature models competition for projects as an auction in which jurisdictions bid against one another through increasingly generous subsidy packages, leading to a race to the bottom and leave the winning jurisdiction with a winner's curse if the subsidy exceeds the local economic benefits generated by the investment. A second strand examines statutory policies adopted at the state or federal level, including geographically differentiated corporate taxes, payroll taxes, employment credits, and broad investment incentives (Gale et al., 2015; Suárez Serrato and Zidar, 2016; Giroud and Rauh, 2019; Rauh and Shyu, 2024; Ku et al., 2019; Neumark and Grijalva, 2017). I contribute to the latter.

Within this second strand, the paper advances research on capital investment tax policy, an important but understudied area of business taxation. Most of this literature studies accelerated depreciation and expensing provisions and finds that reductions in the cost of capital increase investment (Hall and Jorgenson, 1967; Cummins et al., 1994; House and Shapiro, 2008; Edgerton, 2010; Zwick and Mahon, 2017; Maffini et al., 2019; Ohrn, 2019). Some work also finds that these investment responses increase local employment, consistent with capital and labor being complements (Garrett et al., 2020). I contribute to this literature in four ways. First, I study sales tax exemptions on capital purchases and property tax exemptions on installed capital, a relatively understudied form of investment incentive. Second, I show that complementarity between capital and labor does not guarantee an increase in employment across the local economy. Subsidized investment may raise resource costs and displace incumbent firms, reducing local employment overall despite the positive labor demand the project generates. Third, I extend the analysis beyond investment and employment to examine effects on municipal finances and local resource markets. These outcomes are important to study with respect to sales and property tax exemptions because the policies directly affect local revenues while potentially increasing demand for infrastructure and public services. Finally, I build on the literature by studying the exposure municipalities face after an incentive attracts a rapidly growing industry.

The paper therefore adds a dynamic dimension to evaluating place-based incentives.

Local consequences depend not only on whether incentives attract investment, but also on how the targeted industry subsequently grows and changes.

Finally, this paper contributes to a growing literature on the economic consequences of the digital and AI economy. Existing work has primarily emphasized the effects of artificial intelligence on productivity (Brynjolfsson et al., 2023), labor markets (Babina et al., 2023), and aggregate economic growth (Acemoglu, 2024). Artificial intelligence requires substantial physical infrastructure, yet the local economic consequences of this infrastructure remain understudied. Contemporaneous work by Alvarez et al. (2026) and Yue and Zeng (2026) examines local employment and electricity-market effects of data center development. I complement this work by studying a broader set of local outcomes, including municipal borrowing costs, government revenues and expenditures, land prices, and political outcomes. Studying these outcomes is important for understanding the consequences of the expansion of the digital economy.

The remainder of the paper is organized as follows. Section 2 provides institutional background on data centers. Section 3 develops the theoretical framework. Section 4 describes the data. Section 5 presents the empirical methodology, and Section 6 reports the results.

2 Institutional Background

Data centers provide a useful setting for four reasons: they are very capital-intensive, their location is extremely responsive to tax incentives, geographic differences in county suitability create within-state variation in exposure to these policies, and their scale and resource demands have increased rapidly with technological change. These features allow me to causally identify the local effects of investment incentives and how those effects change as the subsidized projects become larger and more resource-intensive. This section lays out relevant institutional details for the empirical setting.

2.1 Prevalence of Data Centers

Data centers are highly capital-intensive and increasingly central to the modern and AI-enabled economy. In the United States, spending on data center construction has risen dramatically over the past decade, making digital infrastructure a key component of aggregate investment. As of January 2026, U.S. spending on data center construction exceeded $2.4 billion per month (Figure A.4), roughly sixteen times the level observed in early 2014 (Samborska, 2026). Globally, estimated demand for data center capacity reached approximately 82 gigawatts in 2025 (McKinsey & Company, 2025a). Data centers house the servers, storage systems, and networking equipment needed to store, process, and transmit large volumes of data. As cloud computing, artificial intelligence, and digital services expand, demand for data center capacity is expected to grow exponentially. Some estimates suggest AI workload demand could increase more than 3.5-fold by 2030 (Figure A.5) (McKinsey & Company, 2025b).

2.2 Data Center Subsidies

Data center incentives generally take three forms: full or partial abatement of state sales taxes, local sales taxes, and local property taxes. State sales tax abatement reduces the taxes paid on qualifying purchases, which, depending on the state, may include servers, cooling equipment, electricity, and construction materials. Local sales tax abatement provides relief for taxes imposed by counties and municipalities on similar purchases. Property tax abatement reduces the recurring taxes paid on the real and personal property, like buildings and installed equipment, associated with a facility. Sales tax abatement therefore subsidizes the purchase of new capital, while property-tax abatement reduces the continuing cost of owning that capital.

By 2025, 40 states had adopted some form of data center tax incentive (Table B.2). All 40 states provide either a full or partial abatement of state sales taxes on qualifying data

center purchases or impose no state sales tax (Table B.4). Similarly, 39 states provide a full or partial abatement of applicable local sales taxes or impose no local sales tax. Property tax abatement are less common: eight states, representing 20 percent of adopters, include them in their data center incentive programs. Figure A.6 illustrates how these incentives reduce the tax liability of a qualifying data center.

Eligibility for the exemptions is conditioned on the scale of the proposed project. Thirty-six states impose a minimum investment threshold, while 29 states impose a minimum employment requirement. The mean investment threshold is $110.8 million, and the mean employment threshold is 17.3 jobs. Wage requirements are considerably less common; only eight states require qualifying projects to meet a wage standard. These conditions highlight how capital incentives differ from policies tied primarily to local employment. Whether a project is eligible or not is generally determined by the amount of capital invested, while employment and wage requirements are less common and relatively modest compared with the scale of the qualifying investment. Moreover, the value of the tax abatement can increase with the amount of equipment purchased and property installed without requiring a proportional increase in employment. Therefore, data center incentives provide a useful setting for examining how subsidies tied to capital investment affect local economies.

These incentives can meaningfully affect where data center investment occurs. For example, one study estimates that approximately 30 percent of data center activity in Georgia may be attributable to the state's tax incentives (Justice and Shepherd, 2025). The staggered adoption of these policies provides exogenous variation in exposure to capital-intensive development as states become more attractive to data center investment.

2.3 Geographical Suitability

The setting offers rich within-state geographic variation. Figure 2 maps the water-adjacent and non-water-adjacent counties. A concern with using state tax incentives is that their

adoption may coincide with other state-level policy changes or shocks. Even when incentives apply statewide, their effect on actual data center investment is concentrated in counties that are suitable for development. In this context, water-adjacent counties are more exposed because large data centers often require access to water resources for cooling. Figure A.7 illustrates this spatial pattern by showing that data centers tend to cluster near major rivers and other water bodies. This creates within-state variation as water-adjacent and non-water-adjacent counties face the same state-level policy environment, but water-adjacent counties are more likely to receive subsidy-induced data center investment. Appendix C provides additional details on other factors that influence data center location.

Appendix C provides additional details on other factors that influence data center location. I leverage the staggered timing of tax incentives and county suitability for data center development to identify the causal effects of data center tax incentives on local economic outcomes.

2.4 Technological Change and Evolution of Data Centers

The scale of data centers has grown dramatically over time. Early facilities were small operations housed in office buildings or industrial parks (Figure A.8). By contrast, contemporary hyperscale data centers that support cloud computing and artificial-intelligence workloads operate on a substantially larger scale. These large facilities often span hundreds of thousands or even millions of square feet and require hundreds of megawatts of electricity. Appendix D provides additional details on the different types of data centers and their operating characteristics.

Major technology firms, such as Amazon, Google, Meta, and Microsoft, have sharply increased capital expenditures directed toward new data center infrastructure. Recent data center announcements illustrate the scale of this shift. For example, Meta's Hyperion data center in Richland Parish, Louisiana, is expected to reach approximately 15 million square

feet and consume 5 gigawatts of power at full capacity, approximately half of New York City's peak electricity consumption. This underscores how data center development has shifted from individual facilities toward massive infrastructure campuses (Goldman, 2026).

Using facility-level data, I document this systemic shift in Table B.5, which reports cumulative data center development growth over the sample period. Figure A.9 plots the corresponding cumulative series. However, these cumulative measures do not show whether the typical facility has become larger.

I therefore examine the evolution of facility scale separately. Table B.5 reports annual median facility square footage and power capacity, and Figure A.10 plots these medians over time. Both measures generally increase over time, with substantially larger and more power-intensive facilities in the post-2016 period. This pattern suggests that the industry expanded along both the extensive and intensive margins: not only were more facilities built, but the typical facility became larger and more power-intensive. The vertical markers highlight major industry developments, including accelerating cloud adoption, the expansion of hyperscale computing, and the emergence of generative-AI demand.

Table B.6 provides a more detailed comparison of facilities before and after 2016. I also formally assess the post-2016 change by regressing logged facility size and power capacity on a post-2016 indicator, with state fixed effects and state-clustered standard errors. Table B.7 reports the estimates and Figure 3 plots year-specific estimates. These results show that later facilities are larger and more power-intensive from 2016 onwards, suggesting that later-adopting states may attract projects with different local consequences.

3 Theoretical Motivation

Ex-ante, the relationship between incentives for capital-intensive projects and economic activity is not obvious. Whether incentives increase data center investment depends on how much they lower the cost of acquiring capital and on whether the investment would have occurred anyway. In the neoclassical model, firms invest until the marginal return from

an additional unit of capital equals its marginal cost. The costs include the price of capital goods, depreciation, financing costs, and taxes paid on them. Hall and Jorgenson (1967) show that tax incentives stimulate investment by lowering the user cost of capital. Similarly, data center incentives reduce the cost of acquiring capital through sales tax exemptions, which reduce the cost of acquiring servers, cooling equipment, and other materials at the time of purchase, and property tax exemptions, which reduce the recurring costs of holding capital. However, House and Shapiro (2008) show that the duration of the subsidies also matters. With temporary incentives, investment responses may partly reflect changes in timing rather than lasting increases in the capital stock, as firms shift capital purchases into the subsidy window to take advantage of the tax benefits. Since data center incentives are standing exemptions, they may instead increase the desired capital stock and generate investment that builds over time.

Whether or not the incentives increase investment is important for understanding the effects of the incentives on public finance outcomes. If we hold investment fixed, tax exemptions reduce the revenue collected from eligible facilities. On the other hand, if incentives attract investment that would otherwise have located elsewhere or not occurred, the expansion of the tax base can generate additional revenues. Data centers may affect several components of local revenue. Firstly, they can increase tax revenues by paying property tax on the buildings and equipment they invest in and also through the local sales tax they pay on construction activity and purchases from suppliers. Greater demand for electricity and water may also increase utility tax receipts and utility revenue.

On one hand, revenues from projects may allow governments to expand education and other public services. On the other hand, these revenues may be spent to provide additional infrastructure like roads and utilities needed to accommodate the projects. Therefore, fiscal competition can shift public spending toward inputs that attract mobile capital and away from public goods consumed by residents.

Ex ante, the effect of incentives for capital-intensive projects on employment is not

obvious. Lower investment costs can encourage firms to expand production and hire more workers, but the employment response depends on how capital and labor are used together. Garrett et al. (2020) find that locations more exposed to reductions in investment costs experience increases in employment and total earnings, but no positive effect on earnings per worker. On the other hand, Ohrn (2019) find that state accelerated depreciation incentives increase manufacturing investment but find no statistically significant employment response. Greenstone et al. (2010) find that large plant openings generate productivity gains for existing plants, particularly those sharing similar labor and technology pools, but also increase local labor costs. Furthermore, Moretti (2010) shows that employment gains from attracting a new business can be partly offset by increases in local wages and prices that raise costs for existing firms. In the data center setting, this mechanism may also operate through electricity and water markets.

Data center expansion creates direct employment and may increase employment in industries that supply the construction and operation of these facilities. At the same time, greater demand for electricity, water, land, and other locally supplied inputs can raise production costs and crowd out activity in other sectors. The model developed in Appendix E formalizes this resource crowding-out mechanism. The model shows that these displacement effects are stronger when data center investment is large relative to local resource capacity. Therefore, total employment gains depend on the extent to which complementary industries gain employment and on employment losses in other industries.

In addition, the model predicts that the effects should differ across industries. Employment should grow more strongly in complementary industries, which, in the case of data centers, include construction and equipment maintenance. Industries that rely heavily on electricity, water, or land but do not benefit from supplying data centers may instead experience weaker employment growth or employment declines.

Finally, the costs may be greater if data center expansion exceeds what policymakers initially expected. An unexpectedly large capacity increase can put additional pressure on

electricity supply, water systems, and land, particularly where infrastructure cannot expand quickly. The resulting costs may exceed the employment and fiscal benefits for some residents, even if the project appeared beneficial when the incentives were introduced. These differences between expected and realized benefits may also have political consequences. If residents experience higher living costs or employment losses and attribute them to the incentives, support for incumbent officials may decline.

4 Data

This section describes the data used for the empirical tests described in Section 5. I first describe the Aterio data, which helps identify data center locations. Next, I describe the data used to study local economic outcomes. Table 1 reports summary statistics, and Table B.1 provides detailed variable definitions.

4.1 Data Center Facilities

Aterio is a commercial data provider that tracks U.S. data center and industrial infrastructure projects. The data provides detailed documentation of when projects are announced, break ground, and become active. For each project, it reports facility details such as location, operators, and financiers. It also provides size measures such as square footage and reported or estimated power capacity. These details are collected and compiled from public filings, permits, company announcements, and satellite imagery. I also use data from Baxtel.com to fill in missing project dates, facility space, and power capacity. This helps improve the completeness of data center information across years, particularly in earlier periods when coverage is more limited. I identify the initiation of a project using the data center's announcement date. If the announcement date is missing, I use the construction-start date, then construction completion, and finally activation, in that order. Because projects are dated using their earliest available development milestone, these

measures capture recorded data center development rather than exclusively completed or operational facilities.

As of 2025, 5,862 facilities have been announced or are operational. I aggregate the facility records to county-years and construct cumulative measures of data center count, facility space in square feet, and power capacity in megawatts. Because projects are recorded at their announcement date, the resulting measures capture data center development rather than only operational facilities. This is particularly important because it allows me to measure economic activity that occurs not only during the operational phase of a data center but also during construction. During the construction phase, projects generate demand for labor, materials, and infrastructure, which may begin to affect local employment, public finances, and resource use even before the facilities become operational.

Panel A of Table 1 reports summary statistics on the constructed variables, using 90,740 county-year observations from 1996–2024. The average county-year has 0.267 recorded data centers, 31,050 square feet of facility space, and 5.358 megawatts of power capacity. The median is zero for all three measures, indicating that recorded development is concentrated in a relatively small share of county-years.

4.2 Local Government Revenues and Expenditures

The U.S. Census Bureau's Annual Survey of State and Local Government Finances provides annual information on government revenue, expenditure, and debt. The Census surveys the full universe of state and local governments in years ending in two and seven, while the intervening annual surveys cover state governments and a sample of local governments. The data combine government questionnaires, administrative records supplied by state agencies, and information compiled from government financial reports.

The survey includes observations from county governments, municipalities, townships, school districts, and special districts. I assign each government to counties using its 5-digit FIPS code and aggregate amounts by county and year. Changes in government coverage

reflect random survey selection rather than systematic changes associated with incentive adoption. Such coverage variation alone does not imply bias in the estimated treatment effects.

Panel B of Table 1 reports summary statistics on government revenue. Average annual county-level revenue is $278.2 million. Tax revenue averages $113.1 million, general charges average $46.2 million, and intergovernmental revenue averages $115.7 million. Within tax revenue, property taxes average $86.9 million and sales taxes average $14.3 million.

Panel C reports summary statistics on government expenditure. Average annual expenditure is $319.9 million. Current operations account for the largest expenditure category, averaging $257.1 million, while combined capital expenditure averages $40.6 million, including $29.9 million in construction spending. By function, education expenditure averages $138.6 million, utilities expenditure averages $33.8 million, and infrastructure expenditure averages $30.3 million.

Panel D reports summary statistics on government debt. Total long-term debt outstanding at year-end averages $305.5 million, with a median of $33.4 million.

4.3 Municipal Bonds

The Municipal Securities Rulemaking Board (MSRB) collects transaction reports submitted by municipal securities dealers through its Real-Time Transaction Reporting System (RTRS) beginning 2005. The records identify securities by CUSIP and report details such as trade dates, prices, yields, par amounts, and transaction types. The system covers customer purchases and sales as well as inter-dealer trades.

I aggregate the observations to the bond-month level by calculating the par-value-weighted average transaction yield. In addition to yields, I also examine the Treasury-adjusted yield spread. This is calculated by subtracting the bond's Treasury-based risk-free benchmark from its municipal yield, using the coupon and maturity information described in Table B.1. Both outcomes are measured in percentage points. These measures capture

the yields investors require in the secondary market. The sample described in Panel H of Table 1 covers 2005–2023. The mean volume-weighted yield in the descriptive sample is 2.52 percent, and the median is 2.41 percent.

4.4 Employment

The Bureau of Labor Statistics' Quarterly Census of Employment and Wages (QCEW) covers approximately 95 percent of U.S. wage and salary employment and publishes establishment counts, monthly employment, and quarterly wages by industry and geography. I use the private-sector county and industry series to examine employment and wage outcomes.

I construct quarterly employment by averaging the monthly employment counts for a quarter. I also construct average quarterly wages by dividing total quarterly payroll by average employment. Establishments are individual operating locations rather than firms, so a firm with several locations may contribute more than one establishment.

For the overall analysis, I use the published county totals, and then conduct sectoral analysis separately using the industry-county panels. I examine industries that complement data centers, such as nonresidential building construction (NAICS 2362), electric power generation and supply (2211), and natural gas distribution (2212), as well as non-complementary industries like residential building construction (2361). Sectoral estimates describe changes among county-industry observations with available positive outcomes. Disclosure restrictions and changes in the observed sample may affect these estimates, particularly in smaller sectors.

Panel E of Table 1 shows the summary statistics and reports an average of approximately 34,813 workers and 2,587 establishments per county-quarter, with mean quarterly wages of $8,228.

4.5 U.S. Energy Information Administration Electricity Data

I use data from the U.S. Energy Information Administration's (EIA) Annual Electric Power Industry Report, Form EIA-861, to measure residential electricity rates. The EIA collects information from electricity distribution utilities and power marketers on retail electricity sales, revenues, and customer counts. The annual files also include service-territory information which helps identify the counties and states in which each power plant distributes electricity. Using this information, I can link utility-level residential sales to counties being served and calculate county rates.in

I construct the residential electricity rate by dividing residential electricity revenue by residential electricity sales, expressed in dollars per kilowatt-hour, and link the rate to counties using the service-territory records. Because suppliers may serve both water-adjacent and inland counties, the estimates capture differential exposure to rate changes. A null estimate does not rule out increases shared across a utility's service territory. The county-year series used in the paper covers 2001–2024. This measure reflects average revenue per unit of electricity sold by the utility serving a county because a utility can serve multiple counties. Panel F of Table 1 reports an average rate of approximately $0.111 per kilowatt-hour.

4.6 Acres.com Land Transaction Data

I use parcel-level land transaction data from Acres.com to measure local land prices. Acres is a commercial land information platform that combines property records, transaction information with geographic characteristics. The database collects information from county courthouse and tax-assessor records. In addition to this, they hand-collect transaction data to provide a complete universe of land transactions. The data provide details on completed transactions, including transaction values, parcel acreage, and location, as well as land-use characteristics used to distinguish different land types. This helps distinguish land used

for development from land used for farming or other agricultural purposes.

The transaction sample used in the paper covers 2010–2025. I aggregate transactions to county-years and calculate price per acre. These measures describe prices of parcels that transact and do not adjust for changes in parcel composition. I also construct this measure specifically for developed land and farmland. Panel G of Table 1 reports summary statistics for these measures.

4.7 American Community Survey Housing Data

The U.S. Census Bureau's American Community Survey (ACS) is a household survey that collects information on demographic, economic, and housing characteristics. I use the five-year estimates, which pool responses over 60 months and provide estimates for all counties and assign each estimate to the ending year of its five-year period. I use median monthly gross rent and median owner-occupied home value to study how data centers affect housing markets.

4.8 Stanford Education Data Archive

The Stanford Education Data Archive (SEDA), produced by Stanford University's Educational Opportunity Project, constructs comparable achievement measures from state mathematics and reading/language arts assessments administered to public-school students in grades 3–8. The underlying information consists of aggregated test-score distributions rather than individual student records. This is particularly helpful because each state uses different assessments and proficiency standards, and this data links these distributions to the National Assessment of Educational Progress (NAEP), placing achievement on a common national scale.

The data is from 2009–2025, excluding unavailable observations in 2020–2021. I aggregate district scores to county-years using subject-specific assessment counts as weights and restrict the geographic match to districts assigned to a single county. This makes sure that

the same district's score is not assigned to multiple counties. Panel I of Table 1 reports summary statistics for the reading and mathematics measures.

4.9 American Local Government Elections Database

I use the American Local Government Elections Database compiled by Justin de Benedictis-Kessner, Diana Da In Lee, Yamil R. Velez, and Christopher Warshaw to examine local political outcomes. The database combines data from state and local government websites, newspaper archives, and other election sources. The data focuses on covering counties with over 50,000 residents. It contains candidate-level vote totals and election outcomes, together with information on incumbency, partisanship, and candidate characteristics. It covers elections from 1989–2021 for mayors, city councils, county executives and legislatures, sheriffs, prosecutors, and school boards.

I match each jurisdiction to a county and use incumbency status and election outcomes to examine whether exposure to data center incentives is associated with incumbents' electoral performance.

5 Empirical Strategy

5.1 Identification Strategy

The empirical analysis aims to estimate the causal impact of data center subsidies on local economic and fiscal outcomes. To study this, I exploit the staggered changes in state-level data center tax incentives (Figure 1). However, there are two main sources of endogeneity that may still exist. Firstly, the states that adopt data center tax incentives may differ systematically from states that do not adopt these policies. Second, even within adopting states, data centers may not randomly locate across counties. Data Centers may choose locations with better infrastructure, lower land costs, or cheap labor. Therefore, comparing counties that receive data centers to those that don't may confound the effects

of data center development with other economic effects.

I address these concerns in a few different ways. First, I show that adopting states are similar to those that do not adopt on various observable characteristics (Figure A.1) and economic trends (Figure A.2). Second, I combine the staggered changes in state-level data center tax incentives with predetermined geographic variation in counties' suitability for data center development and access to water. Data centers can heat up substantially and require large amounts of water for cooling. Being in close proximity to water can help facilitate cooling. Water-adjacent counties are therefore more suitable for hosting data centers as opposed to inland counties. Consistent with this, data centers locate disproportionately in counties with access to major water bodies (Figure A.7).

I create an indicator for water-adjacent counties using county boundary shapefiles from the U.S. Census Bureau and spatial data on major water bodies. I match county polygon to the water bodies using geographic information system (GIS) methods. A county is classified as a water-adjacent county if its boundary intersects a major coastline, lake, river, or other large water body.

Because states adopt data center incentives in different years, I implement the analysis using a stacked difference-in-difference-in-differences framework, following Gormley and Matsa (2011) and Cengiz et al. (2019). The first difference compares outcomes before and after the adoption of a state data center incentive. The second compares adopting states with states that have not yet adopted or never adopt an incentive. The third compares water-adjacent counties with inland counties. The identifying variation comes from changes in outcomes for water-adjacent counties relative to inland counties following subsidy adoption in treated states, relative to the corresponding within-state difference in untreated states.

Conventional two-way fixed-effects estimators can produce difficult-to-interpret estimates when treatment timing is staggered, and treatment effects are heterogeneous across cohorts or over time (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfoeuille, 2020;

Sun and Abraham, 2021; Callaway and Sant'Anna, 2021). The stacked design addresses these concerns by constructing a separate cohort around each treatment event and estimating treatment effects over a common event-time window.

For each year in which a state adopts a data center incentive, I construct a cohort consisting of the states treated in that year and a set of control states that remain untreated over the relevant event window. I drop states that have already adopted a data center incentive from the cohort. The control states are those that are never treated or not yet treated. The cohort-specific samples are then stacked into a single estimation sample. The fixed effects are interacted with cohort indicators, allowing each cohort to have its own set of fixed effects. The baseline specification is:

$$Y_{isct} = \beta \left(Treat_{sc} \times Post_{sct} \times WaterAdjacent_i \right) + \alpha_{ic} + \lambda_{stc} + \mu_{water_i \times t \times c} + \varepsilon_{isct}, \qquad (1)$$

where Y_{isct} denotes an outcome for county i, located in state s, in year t and cohort c. $Treat_{sc}$ equals one for states belonging to the treated group in cohort c, and $Post_{sct}$ equals one following adoption of the data center incentive. $WaterAdjacent_i$ is an indicator equal to one for counties located adjacent to a major water body. The coefficient of interest is β, which captures the incremental effect of subsidy adoption in counties that are more geographically suitable for data center development.

I include county-by-cohort fixed effects, α_{ic}, which absorb time-invariant county characteristics within each stacked cohort. This includes characteristics like geography, baseline economic conditions, and persistent infrastructure differences. I also include state-by-year-by-cohort fixed effects, λ_{stc}, which absorb shocks affecting all counties within a state in a given year, such as changes in state economic conditions or political conditions. I additionally include water-adjacent-by-year-by-cohort fixed effects, $\mu_{water_i \times t \times c}$, which absorb time-varying shocks common to water-adjacent counties.Standard errors are clustered at the state level because treatment is assigned through state-level policy adoption and shocks

may be correlated across counties within a state and over time.

The identifying assumption is that, absent the adoption of data center incentives, the difference in outcomes between water-adjacent and inland counties in treated states would have evolved similarly to the corresponding difference in untreated states. The event-study specification below examines whether differential trends emerge prior to subsidy adoption.

5.2 Event Study Specification

To examine treatment dynamics and assess pre-treatment trends, I replace the post-treatment indicator in Equation 1 with indicators for event time:

$$\begin{aligned} Y_{isct} = &\sum_{k \neq -1} \beta_k \left[Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}(t - T_{sc} = k) \right] \\ &+ \alpha_{ic} + \lambda_{stc} + \mu_{water_i \times t \times c} + \varepsilon_{isct}, \end{aligned} \tag{2}$$

where T_{sc} denotes the treatment year for cohort c, and $\mathbf{1}(t - T_{sc} = k)$ is an indicator equal to one when an observation is k years relative to subsidy adoption. Event time $k = -1$ is omitted and serves as the reference period.

The coefficients β_k show the differential evolution of outcomes in water-adjacent counties in treated states relative to the corresponding water-adjacent–inland difference in untreated states. Pre-treatment estimates provide a test for differential trends before subsidy adoption, while post-treatment estimates show when the effects emerge and how they evolve over time. The event-study specification includes the same fixed effects and state-level clustering as the baseline model.

6 Results

6.1 Data Centers

I start by examining whether the empirical design captures increases in data center development activity, as this initial analysis is important for interpreting the subsequent results. I show that incentives attract investment to geographically suitable areas, i.e. those that are adjacent to water bodies.

Table 2 reports the results. Column 1 examines the change in cumulative data center count, while Columns 2 and 3 examine cumulative facility space and power capacity. These outcomes provide a complete picture of how data center development increases after incentive adoption. Count measures the number of projects, whereas space and power capacity provide insight into aggregate physical scale. The regressions use county-year observations from 1996 through 2024. Each outcome — data center counts, square footage, and power capacity — uses its own average and is scaled by its overall sample average. For example, if average facility space is 10,000 square feet, a county with 20,000 square feet gets a normalized value of 2. A regression coefficient of 0.3 then corresponds to 30% of the sample average—not necessarily 30% growth in that county. The means are reported in Panel A of Table 1.

Column 1 shows an increase of 0.303 ($p < 0.05$) in data center count following incentive adoption relative to control counties, corresponding to an increase of 30.3 percent of the normalization mean. With a normalization mean of 0.267 facilities, this represents approximately 0.081 additional facilities in more exposed counties relative to the comparison group. Since data centers are discrete and relatively uncommon projects, this estimate describes an average response across counties rather than an increase of the same size in every county.

Columns 2 and 3 show that the increase in project count is accompanied by increases in facility space and power capacity. The estimates equal 42.3 percent ($p < 0.1$) of the

normalization mean for facility space and 50.2 percent ($p < 0.05$) of the normalization mean for power capacity, corresponding to approximately 13,125 additional square feet and 2.69 MW of additional capacity. These results further suggest that incentives attract data centers in water-adjacent counties and increase the aggregate physical scale. The power-capacity response is large relative to its normalization mean, which highlights that electricity demand is an important margin to examine in the subsequent analysis.

Figure 4 presents the annual event study for data center count using county-year observations from 1996 through 2024. Event year 0 is the year of incentive adoption, and year -1 is the reference period. The joint test that the coefficients for event years -5 through -2 equal zero is not rejected ($p = 0.284$), which supports the parallel trends assumption. Since I use announcements of projects under construction, these estimates capture development activity rather than only operating facilities.

Taken together, the results indicate that incentive adoption is followed by an expansion in the number and aggregate capacity of data center projects in geographically suitable counties.

6.2 Local Public Finances

I examine how data center tax incentives affect local government finances. The estimates compare changes in water-adjacent and nonwater-adjacent counties following state incentive adoption with the corresponding differences in comparison states. The specifications include county, state-by-year, and water-by-year fixed effects, each interacted with cohort. I estimate the financial outcomes using Poisson pseudo-maximum likelihood to accommodate outcomes with value 0.

The fiscal consequences of incentives for data centers are not obvious ex ante. While data center projects may expand the local property and commercial tax base, the incentives used to attract them reduce taxes on qualifying investment and activity. In addition to this electricity, water, and infrastructure requirements may also increase public spending.

To explore this further, I examine whether revenue gains are accompanied by spending on services and amenities for residents, and how debt issuance changes after incentives are adopted.

6.2.1 Revenue

Panel A of Table 3 reports the effects on total revenue and its components. Following incentive adoption, water-adjacent counties in adopting states experience increases in total revenue (Column 1; $p < 0.05$) and total taxes (Column 2; $p < 0.05$) relative to the comparison counties. The estimates imply increases of approximately 2.0 and 2.5 percent, respectively. For an average county with mean revenue of $278.2 million and tax revenue of $113.1 million (Table 1), this implies an additional $5.7 million in annual revenue and $2.8 million in annual tax receipts. I find a positive change in general charges and intergovernmental revenue, but neither is statistically significant (Columns 3 and 4).

Panel B examines the tax categories underlying this result. General and other selective sales tax receipts increase by approximately 5.5 percent (Column 1; $p < 0.05$), and property tax receipts rise by 1.9 percent (Column 2; $p < 0.10$). I do not find statistically significant changes in public utility taxes or license taxes (Columns 3 and 4). The increases in sales and property tax receipts suggest that incentives can increase revenues by expanding the tax base, despite having exemptions on qualifying purchases and property tax.

In Panel C, I examine various components of general charges, which are the revenues from government services. I find that revenues from publicly operated utilities increase by approximately 4.7 percent (Column 1; $p < 0.05$). This measure captures cash flows from public water, electricity, gas, and transit systems and is different from the utility taxes examined in Panel B. This increase is consistent with greater demand for publicly provided utilities following incentive adoption. I also examine revenues from other government services and do not find statistically significant revenues from parks and recreation, education, or sewerage charges (Columns 2–4). Figure 5 examines the timing of these effects.

6.2.2 Expenditure

The revenue gains are accompanied by higher public spending. However, it is important to understand what the expenditure is on. Table 4 examines aggregate expenditure, capital investment, public services, and education outcomes.

Panel A shows that total expenditure increases by approximately 3.1 percent (Column 1; $p < 0.05$). Direct expenditure, which excludes transfers to other governments, rises by 3.0 percent (Column 2; $p < 0.05$), while current operating expenditure increases by 2.1 percent (Column 3; $p < 0.05$). Intergovernmental expenditure does not change significantly (Column 4).

Panel B shows that the spending response also includes investment in physical assets. Combined capital expenditure, which includes construction spending and purchase of other capital, increases by approximately 5.6 percent (Column 1; $p < 0.05$). Specifically, construction spending rises by 5.5 percent (Column 2; $p < 0.10$) and other capital outlays, including purchases such as equipment and land, increase by 8.8 percent (Column 3; $p < 0.05$). I also find that interest expenditure does not change significantly (Column 4).

Panel C provides more direct evidence on the purposes of this spending. Utilities and infrastructure expenditure do not change significantly (Columns 1 and 2), while services expenditure increases by approximately 2.3 percent (Column 3; $p < 0.05$). This category includes education, parks, libraries, public safety, health, welfare, and housing. Assistance and subsidies also increase by approximately 21.6 percent (Column 4; $p < 0.01$).[7] The increase in services expenditure suggests that the expenditure supports services and amenities for residents.

Panel D examines education expenditure in greater detail. Total education expenditure increases by approximately 1.8 percent (Column 1; $p < 0.05$), while education construction expenditure rises by 8.2 percent (Column 2; $p < 0.10$), suggesting that broader increases in public spending include investment in educational facilities. I also examine student

[7]The assistance and subsidies regression uses 42,024 stacked observations, compared with 271,637 for total expenditure.

achievement using county averages of Stanford Education Data Archive scores. Math scores increase by approximately 0.012 standard deviations (Column 4; $p < 0.05$), while reading scores do not change significantly (Column 3). These results provide evidence of improved math achievement alongside higher education spending.

Figure 6 presents the corresponding event studies for total expenditure, current operations, combined capital expenditure, and services expenditure. Total and operating expenditure estimates generally become more positive in the years following adoption. Capital and services expenditure also show larger positive estimates later in the post period, consistent with spending adjustments developing over time.

Together, the expenditure results suggest that the revenue gains are accompanied by spending on public services and facilities that ultimately serve residents.

6.2.3 Debt and Borrowing Costs

,I next examine whether the increases in revenue and expenditure are accompanied by changes in outstanding debt and borrowing costs. Table 5 shows no statistically significant change in total long-term debt outstanding (Column 1). I therefore examine its composition, distinguishing debt issued for private purposes from debt for government purposes. Private-purpose debt finances private activities through government borrowing arrangements, whereas government-purpose debt funds projects undertaken by the local government.

Debt outstanding in the government purpose category increases by approximately 5.5 percent (Column 2; $p < 0.01$). The private-purpose estimate is negative but statistically insignificant (Column 3), and there is no statistically significant change in short-term debt outstanding (Column 4). These estimates suggest an increase in borrowing within the government purpose category alongside little change in total long-term debt. Together with the expenditure results, this pattern is consistent with greater government activity and spending on public services and facilities.

Municipal borrowing costs also increase. Bond yields rise by 11 basis points (Column 5; $p < 0.05$), while Treasury-adjusted spreads increase by 10.4 basis points (Column 6; $p < 0.10$). The increase in spreads indicates that higher borrowing costs are not explained solely by changes in Treasury rates. It suggests that investors require additional compensation for holding municipal debt in the affected counties, potentially reflecting concerns about future spending commitments or reliance on a more concentrated local economic base. It may also reflect changes in the mix of traded bonds. The estimates do not isolate changes in borrowing costs for otherwise comparable bonds.

Taken together, the fiscal results suggest that data center incentives expand local revenues and support greater spending on services and facilities used by residents. These changes occur without a statistically significant increase in total long-term debt outstanding, but they are accompanied by higher municipal borrowing costs.

6.3 Employment

Employment growth is one of the primary arguments for offering investment incentives. Data center construction and operation may create jobs in construction, utilities, and related services. At the same time, these facilities require relatively few workers compared with the scale of their investment. Their demand for electricity, land, water, and infrastructure may also raise operating costs for businesses that compete for these resources.

Table 6 Panel A reports the employment estimates. The number of establishments declines by approximately 0.75 percent(Column 2; $p < 0.05$). Total private employment declines by approximately 1.0 percent (Column 1; $p < 0.1$), and average quarterly wages decline by approximately 0.53 percent (Column 3; $p < 0.05$). Figure 7 presents the corresponding employment event study. These results suggest that the increase in data center investment does not translate into broader employment or wage gains.

The sectoral estimates show where employment gains occur. Panels B–E of Table 6 report the results by industry. Electric power employment increases by approximately 8.7

percent, while wages increase by 1.9 percent (Panel B, Columns 1 and 3; both $p < 0.10$). Natural gas employment and wages increase by approximately 86.2 percent and 22.9 percent, respectively (Panel C, Columns 1 and 3; both $p < 0.01$), although there are significantly fewer observations for this industry we should caustion while interpreting these results. Employment in HVAC and semiconductor manufacturing increases by approximately 11.0 percent (Panel D, Column 1; $p < 0.10$). In contrast, residential construction employment declines by approximately 3.8 percent (Panel E, Column 1; $p < 0.01$). This pattern suggests that data center development increases demand for workers in industries that support these projects even as aggregate employment falls.

I next examine whether the employment decline is concentrated in industries that rely more heavily on local resources. Table 7 reports the effects of data center incentives on employment, establishments, and wages across complementary, resource-intensive, and other industries. Panel A shows that employment in complementary industries increases by approximately 6.1 percent (Column 1; $p < 0.01$). The estimated effects on establishments and wages are positive but not statistically significant.

Panel B reports the results for resource-intensive industries. Establishments decline by approximately 1.0 percent (Column 2; $p < 0.01$), while wages decline by approximately 0.34 percent (Column 3; $p < 0.10$). The employment coefficient is negative but not statistically significant.

These results suggest that data center incentives increase employment in industries that support data center activity, while resource-intensive industries experience declines in establishments and wages. The pattern is consistent with the the crowding out mechanism.

Finally, Table B.8 examines whether these employment changes are accompanied by changes in the number of residents or migration patterns. I find no statistically significant change in population or total net migration per 1,000 residents through 2022 (Panel A), with similar results through 2023 (Panel B).

6.4 Resource Prices

Data centers require substantial electricity capacity and large development sites. As more data centers come online and increase demand for these inputs, prices may rise. However, the effect on residential electricity prices depends on how utilities allocate the costs of additional generation and grid investment. There may also be heterogeneity in how land prices react, depending on the type of land and its use.

To explore these effects, I first examine the prices of electricity (Table 8). I find that prices increase by approximately 0.2 percent, but the estimate is not statistically significant (Column 1). I also examine how land prices change based on land type. Developed land prices increase by approximately 19.8 percent (Column 2; $p < 0.05$). This is consistent with greater demand for sites suitable for development. On the other hand, agricultural land prices decline by approximately 0.8 percent. Although this estimate is not statistically significant (Column 3), the direction suggests that competition for water may drive down the prices of agricultural land as they struggle to keep up.

6.5 Housing

Data center development may affect housing costs through two competing channels. Greater demand for land may raise land prices and reduce the land available for residential construction, limiting housing supply and increasing house prices and rents. At the same time, noise and pollution associated with data center facilities may make nearby areas less desirable places to live, reducing housing demand and lowering prices.

Table 9 reports the estimates for median monthly gross rent and median owner-occupied home value using county-level American Community Survey data through 2022. I find that median gross rent declines by approximately 0.8 percent (Column 1; $p < 0.05$), while median home value declines by approximately 0.9 percent (Column 2; $p < 0.10$).

These findings suggest that downward pressure on housing demand due to noise and

pollution concerns outweighs any upward pressure from competition for land. The contrast between rising developed land prices and falling residential values suggests that data center development can increase the value of land for development while reducing its appeal for residential use.

6.6 Heterogeneity Across Data Center Eras

The scale and operating requirements of data centers have changed considerably over time. Table B.5 and Figure A.9 document the growth in facility size and power capacity, while Table B.7 shows that facilities with inventory events in 2016 or later are substantially larger and have greater power capacity. Larger facilities may place greater demands on local electricity systems and land, changing how data center investment affects residents and businesses.

To examine these differences, Table 10 reports estimates separately for observations before 2016 and from 2016 onward. Panel A shows no statistically significant change in residential electricity prices or agricultural land prices in the earlier period (Columns 1 and 3). Developed land prices increase by approximately 60.5 percent (Column 2; $p < 0.10$). These estimates suggest that the earlier period is characterized primarily by increased demand for developed sites.

Panel B shows broader changes in resource prices from 2016 onward. Residential electricity prices increase by approximately 0.8 percent (Column 1; $p < 0.10$), and developed land prices increase by approximately 42.6 percent (Column 2; $p < 0.05$). Agricultural land prices decline by approximately 19.2 percent (Column 3; $p < 0.05$). This pattern is consistent with greater demand for developed sites alongside rising electricity costs and declining agricultural land values during the period of larger data center projects. Figure 8 presents the electricity-rate and land-price event studies.

Employment outcomes also differ across the two periods. The earlier employment estimate is small and statistically insignificant (Panel A, Column 4). From 2016 onward,

employment declines by approximately 1.1 percent (Panel B, Column 4; $p < 0.10$). Together with the resource-price results, this suggests that the later period combines greater pressure on local resources with declining aggregate employment.

I also examine local political outcomes. Before 2016, the probability that an incumbent loses office falls by 4.2 percentage points (Panel A, Column 5; $p < 0.05$). From 2016 onward, it rises by 7.6 percentage points (Panel B, Column 5; $p < 0.01$). Incumbents therefore experience improved electoral outcomes in the earlier period and worse outcomes in the later period. This reversal is consistent with residents responding differently as the local costs of data center development become more pronounced.

Taken together, the estimates suggest that the local consequences of data center investment change as facilities become larger and require more resources. The later period is associated with higher residential electricity prices, lower agricultural land values, declining employment, and worse electoral outcomes for incumbents.

6.7 Robustness

Table B.9 in the appendix reports additional checks using shorter event windows, alternative state exclusions, state and year clustering, and exclusions of the largest counties. government-purpose

6.7.1 Lobbying

A potential concern is that the adoption of data center tax incentives reflects firms' existing plans to locate in a state. Firms may choose a location because of favorable economic conditions and then lobby for subsidies. If these conditions also affect local economic outcomes, the estimated effects could reflect these underlying changes rather than the incentives themselves.

To address this concern, I hand-collect evidence of lobbying surrounding subsidy adoption from news reports and local government meeting records. I identify and exclude eleven

states with documented evidence of firm lobbying. Table 11, Panel A, reports the results. The increase in data center count is equivalent to 20.4 percent of its sample mean (Column 1; $p = 0.102$). Total revenue increases by approximately 2.4 percent (Column 2; $p < 0.10$), total taxes by 3.4 percent (Column 3; $p < 0.05$), and total expenditure by 3.4 percent (Column 4; $p < 0.10$). Employment declines by approximately 0.3 percent (Column 5; $p = 0.425$). These findings are consistent with the main results and suggest that they are not driven by states with documented firm lobbying.

6.7.2 Broader Incentive Packages

A second concern is that data center subsidies may be adopted as part of broader incentive packages. Other provisions in these packages could affect local investment, employment, and government finances, making it difficult to distinguish their effects from those of data center incentives.

To address this concern, I examine the legislation and timing of each state's subsidy adoption. I exclude thirteen states where data center incentives were enacted as part of a broader package, retaining states with standalone legislation or no identified broader package. Table 11, Panel B, reports the results. The increase in data center count is equivalent to 30.8 percent of its sample mean (Column 1; $p < 0.05$). Total revenue increases by approximately 2.4 percent (Column 2; $p < 0.10$), and total expenditure increases by 4.7 percent (Column 4; $p < 0.01$). Employment declines by approximately 1.3 percent (Column 5; $p = 0.130$). The total tax estimate remains positive but is not statistically significant (Column 3). The results for investment, revenue, expenditure, and employment therefore persist after excluding states with broader incentive packages.

6.7.3 Local Government Authority

Local governments may also offer their own subsidies to attract data centers. If these incentives coincide with state subsidy adoption and disproportionately benefit water-adjacent

counties, they could contribute to the estimated effects. Tracking individual local incentives is difficult because disclosure is incomplete and negotiations may involve nondisclosure agreements.

To address this concern, I use differences in local government authority under Home Rule and Dillon's Rule. Home Rule generally gives local governments broader discretion over local affairs. Dillon's Rule limits their authority to powers granted by the state or necessarily implied by those powers, requiring local incentives to rest on state authorization. Using the National Association of Counties' classifications, I restrict the sample to fourteen states whose counties operate under Dillon's Rule.

Table 11, Panel C, reports the results. The increase in data center count is equivalent to 51.2 percent of its sample mean (Column 1; $p < 0.10$). Total revenue increases by approximately 3.6 percent (Column 2; $p < 0.05$), total taxes by 5.0 percent (Column 3; $p < 0.01$), and total expenditure by 5.3 percent (Column 4; $p < 0.01$). Employment declines by approximately 2.0 percent (Column 5; $p < 0.01$). These findings suggest that the positive investment and fiscal effects persist in states with more limited local government autonomy.

7 Conclusion

This paper studies the local consequences of state incentives for data center investment and how those consequences change as the underlying technology evolves. Exploiting the staggered adoption of state-level data center incentives and predetermined geographic variation in counties' suitability for data center development, I implement a stacked difference-in-difference-in-differences design.

Data centers provide an especially useful setting to asnswer these questions because they are highly capitla intensive and the industry has undergone rapid technological change while state incentive programs have remained persistent. The findings suggest that policy design should account explicitly for this possibility.

In terms of public finances, data centers increase revenues for local governments. This is primarily through two channels. Firstly, they expand the tax base and increase tax revenue through paying certain sales and property taxes. And second, they increase government services revenue through usage of municipal utilities. In addition to this, data center incentives have meaningful effects on local resource markets, economic activity, and political outcomes, but these effects are not constant over time. In the pre-cloud adoption period, there is an increase developed land values, with comparatively little evidence of pressure on electricity prices, agricultural land, or local employment. After the the adoption of the cloud, when data centers become substantially large, the pattern changes substantially. Data center exposure is associated with higher residential electricity prices, lower agricultural land values, weaker employment, and greater electoral losses for incumbent politicians.

Early adopters design policies when the technology is in its nascent stages, and lae adopters pass relatively similar incentives while receiving substantially different projects. Therefore the incentives apply to projects large in scale, electricity requirements, land demands, and capital intensity of qualifying projects increased. As a result, a policy that initially targeted relatively modest investments could ultimately subsidize projects with substantially different local consequences.

Periodic review, expenditure caps, sunset provisions, and eligibility criteria that adjust to changes in project scale or resource intensity may help prevent incentives designed for one technological environment from producing unintended consequences in another. More generally, the paper shows that the effects of economic development policy cannot be understood independently of the technology being subsidized and that rapidly changing technologies may require equally adaptive policy.

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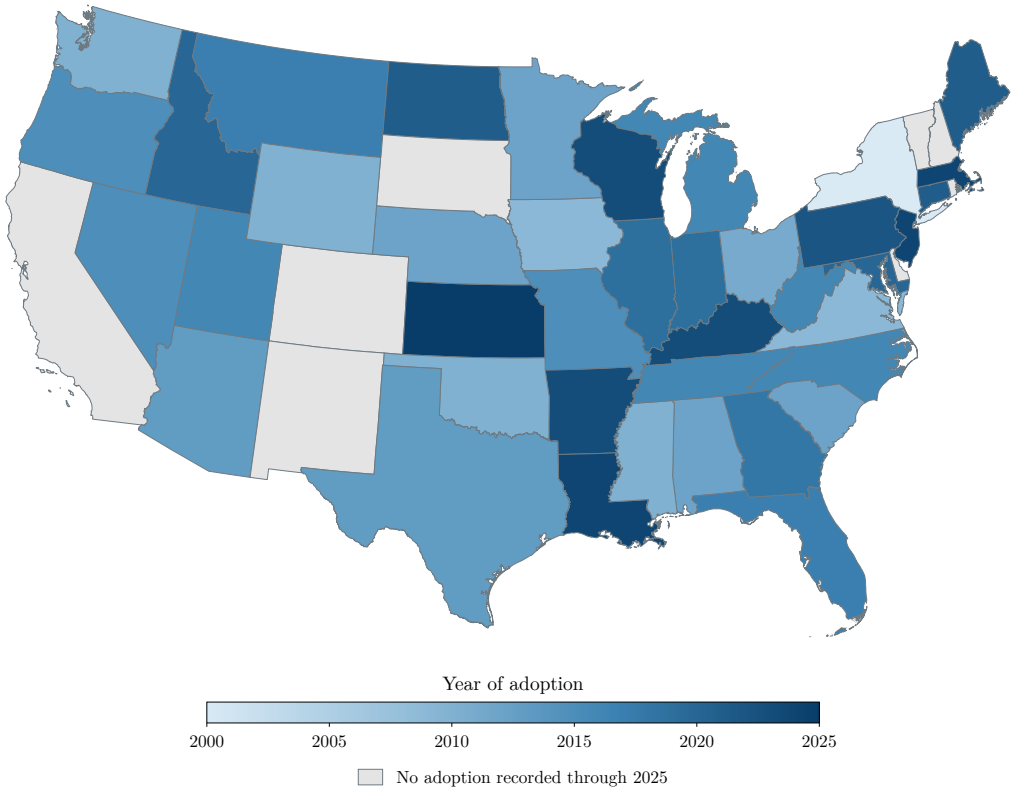
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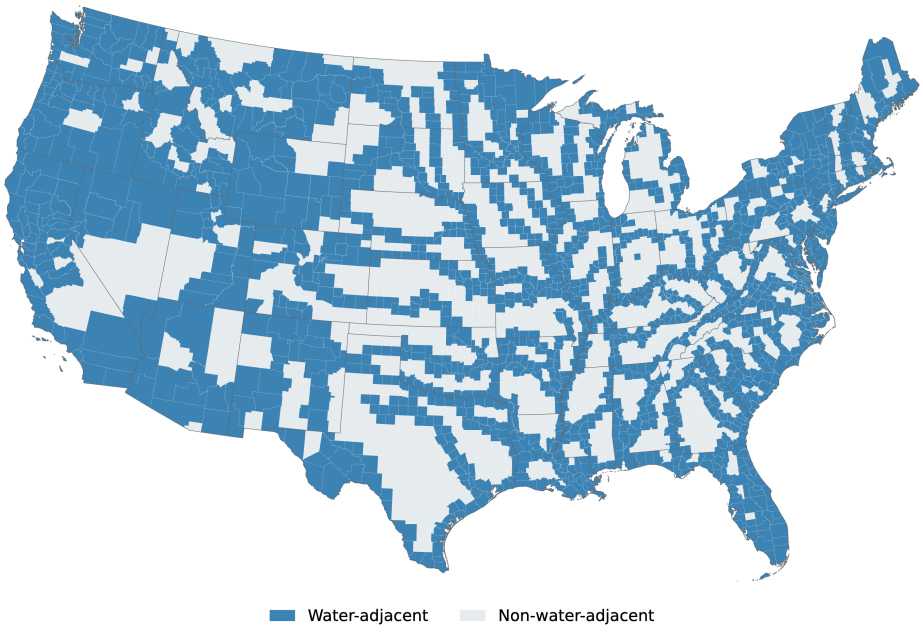
8 Figures

Figure 1. State Adoption of Data Center Tax Incentives



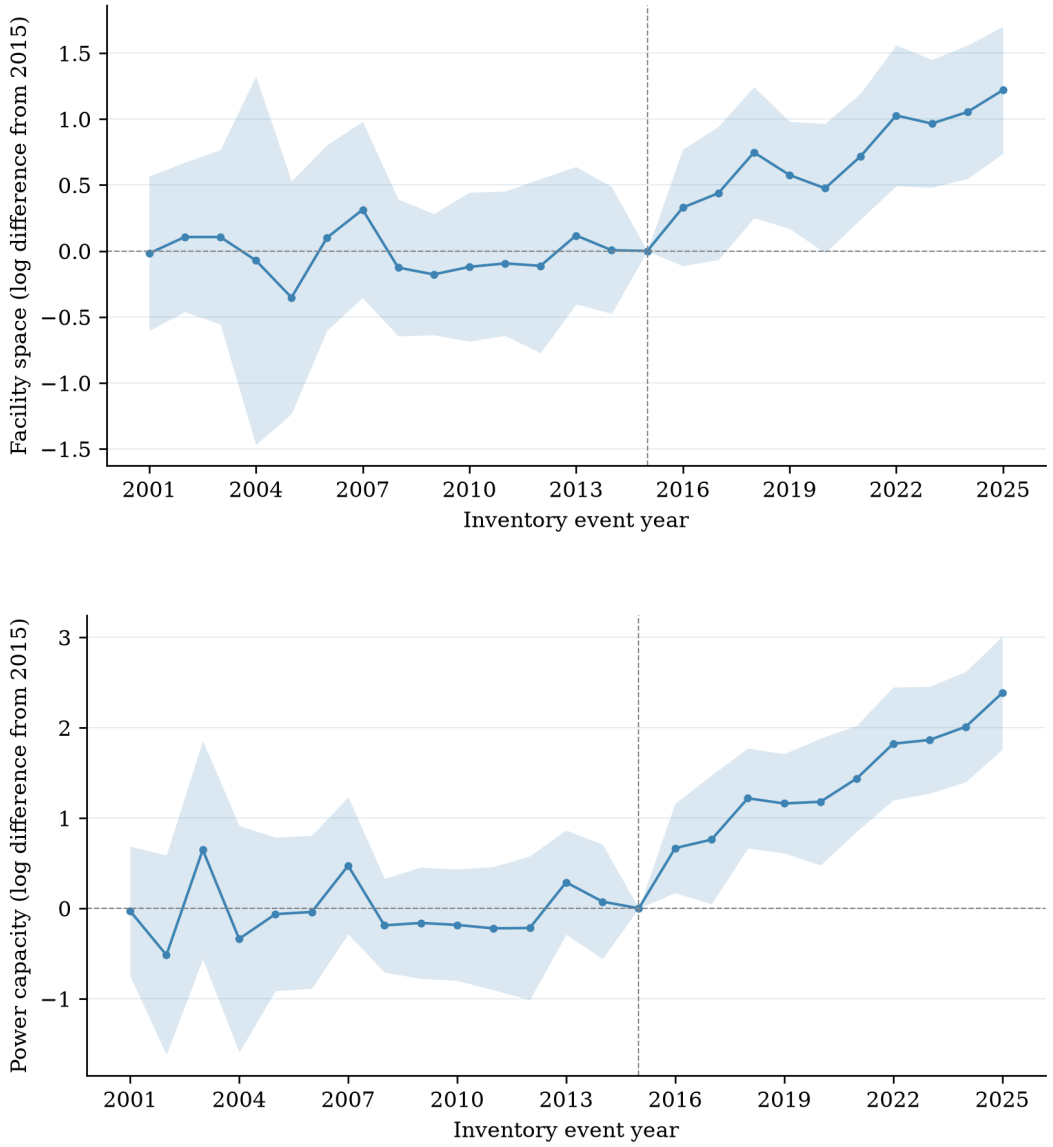
Notes: This figure reports the year in which each state in the United States adopted data center tax incentives, covering adoptions from 2000 through 2025. New York is the earliest adopter in the sample and adopted incentives in 2000. Kansas is the most recent adopter and adopted incentives in 2025. Adoption years are compiled from state legislation and government sources and correspond to those reported in Table B.2. Lighter shades indicate earlier adoption, while darker shades indicate later adoption. Gray identifies states with no recorded adoption through 2025. These adoption years define $Treat_{sc}$, which is an indicator variable that equals one for counties in states that adopt an incentive in cohort c's adoption year. $Post_{sct}$ is an indicator variable which equals one in years following adoption.Figures A.1 and A.2 assess whether adopting states (blue) differ systematically from never-adopting states (gray).

Figure 2. Water-Adjacent and Non-Water-Adjacent Counties



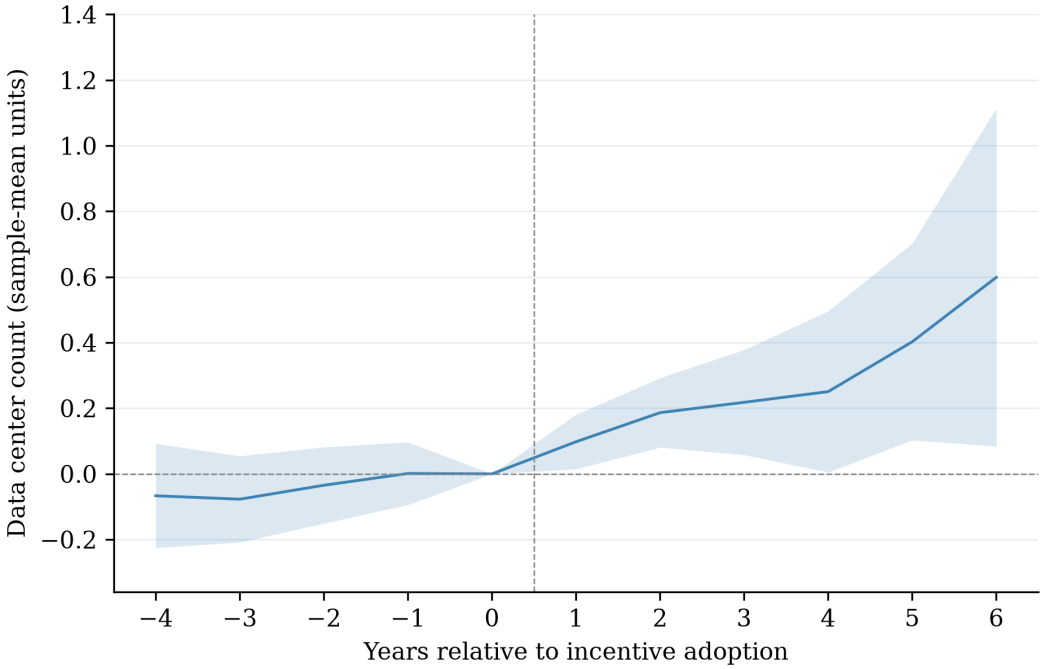
Notes: This figure shows the county water-adjacent classification used in the analysis. Blue identifies water-adjacent counties and light gray identifies non-water-adjacent counties. County boundaries are from the U.S. Census Bureau's 2024 cartographic boundary files.

Figure 3. Evolution of Data Center Scale Post Cloud Adoption in 2016



Notes: This figure shows how data center size and power capacity evolve relative to data centers in 2015. The upper panel reports coefficients from a regression of log facility size measured in square footage. The lower panel reports coefficients for log power capacity measured in megawatts. Both regressions include state fixed effects. The sample includes facilities from 2000 through 2025. Event dates use the announcement date when available, followed by construction start, construction completion, and activation when preceding dates are missing. The omitted reference year is 2015. The shaded area represents the 95 percent confidence intervals. Facility data are from the Aterio U.S. inventory and supplemented with information hand-collected from Baxtel.com. Standard errors are clustered at the state level.

Figure 4. Effects of Data Center Tax Incentives on Data Center Count

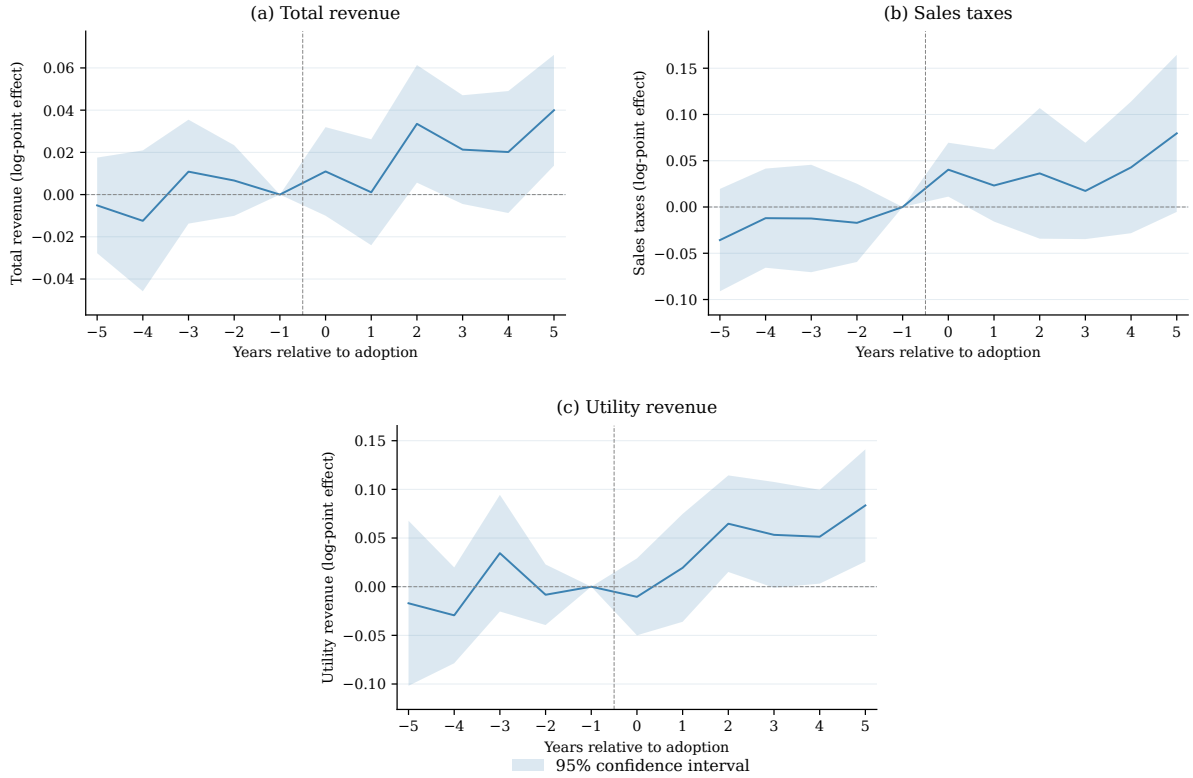


Notes: This figure presents dynamic effects on cumulative data center counts, normalized by the mean across unique county-years from 1996 through 2024. The figure plots the estimated coefficients (β_k) and the shaded 95% confidence intervals from the following stacked triple-difference regression:

$$Y_{isct} = \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}) + \alpha_{ic} + \lambda_{stc} + \mu_{water_i \times t \times c} + \varepsilon_{isct}.$$

hand-collectedY_{isct} represents the cumulative number of data centers in county i and year t in cohort stack c normalized by the 1996–2024 mean. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed, and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption. Event year 0 is the year of adoption. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), state-by-year-by-cohort fixed effects (λ_{stc}), and wateradjacent-by-year-by-cohort fixed effects ($\mu_{water_i \times t \times c}$). The event window spans five years before and five years after the first post year. The data uses county-year observations from 1996–2024. The joint test that the coefficients for event years -5 through -2 equal zero is not rejected ($p = 0.284$). The corresponding pooled estimate is reported in Column 1 of Table 2. Facility data are from the Aterio U.S. inventory and supplemented with information hand collected from Baxtel.com. Standard errors are clustered at the state level.

Figure 5. Effects of Data Center Tax Incentives on Government Revenue

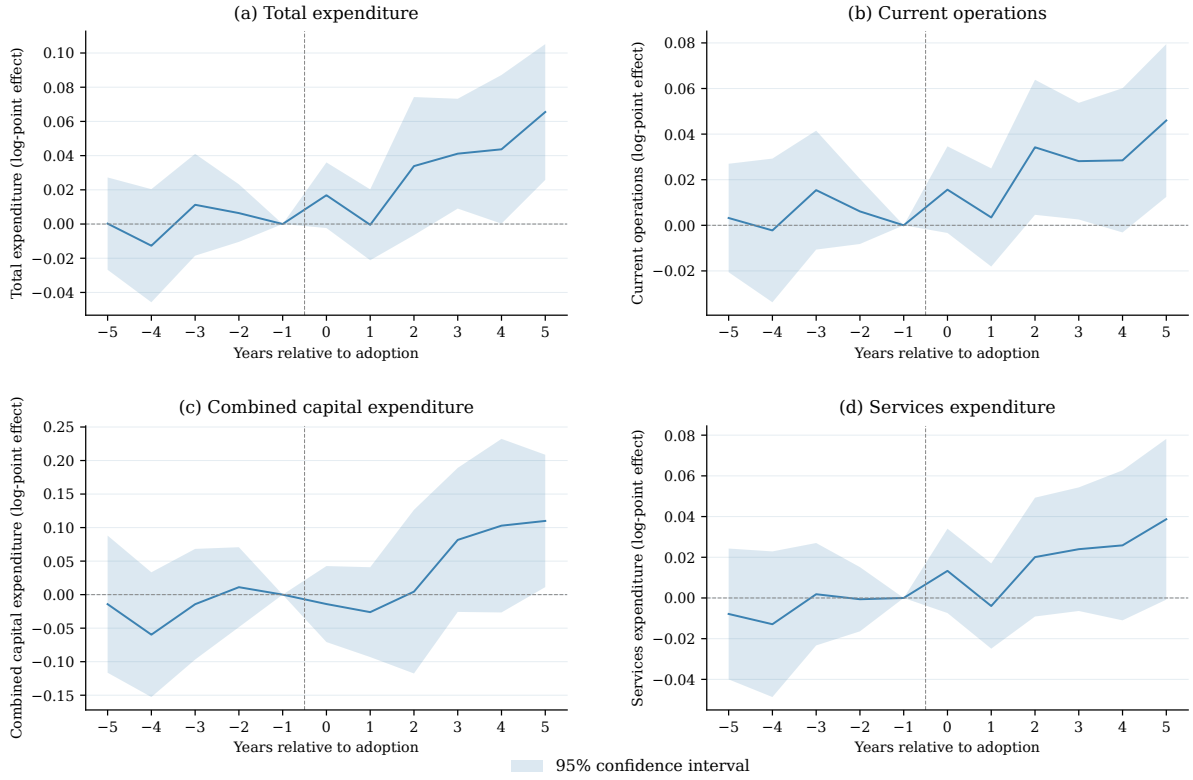


Notes: This figure presents dynamic effects on total revenue (Panel a), sales tax revenue (Panel b), and utility revenue (Panel c). The figure plots the estimated coefficients (β_k) and the 95% confidence intervals from the following stacked triple-difference regression estimated using PPML:

$$\mathbb{E}[Y_{isct} \mid X] = \exp\left[\sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}) + \alpha_{ic} + \lambda_{stc} + \mu_{water_i \times t \times c}\right].$$

Y_{isct} represents the revenue outcome in county i and year t in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed, and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption. Event year 0 is the year of adoption. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), state-by-year-by-cohort fixed effects (λ_{stc}), and wateradjacent-by-year-by-cohort fixed effects ($\mu_{water_i \times t \times c}$). The event window spans five years before and five years after the first post year. The data uses county-year observations from 1996-2022. The joint test that the coefficients for event years -5 through -2 equal zero is not rejected in any panel ($p = 0.623$, 0.656, and 0.425 for Panels (a)–(c), respectively). The corresponding pooled estimates are reported in Table 3. Government revenue data is collected from the Census Annual Survey of State and Local of State and Local Governments. Standard errors are clustered at the state level.

Figure 6. Effects of Data Center Tax Incentives on Government Expenditure

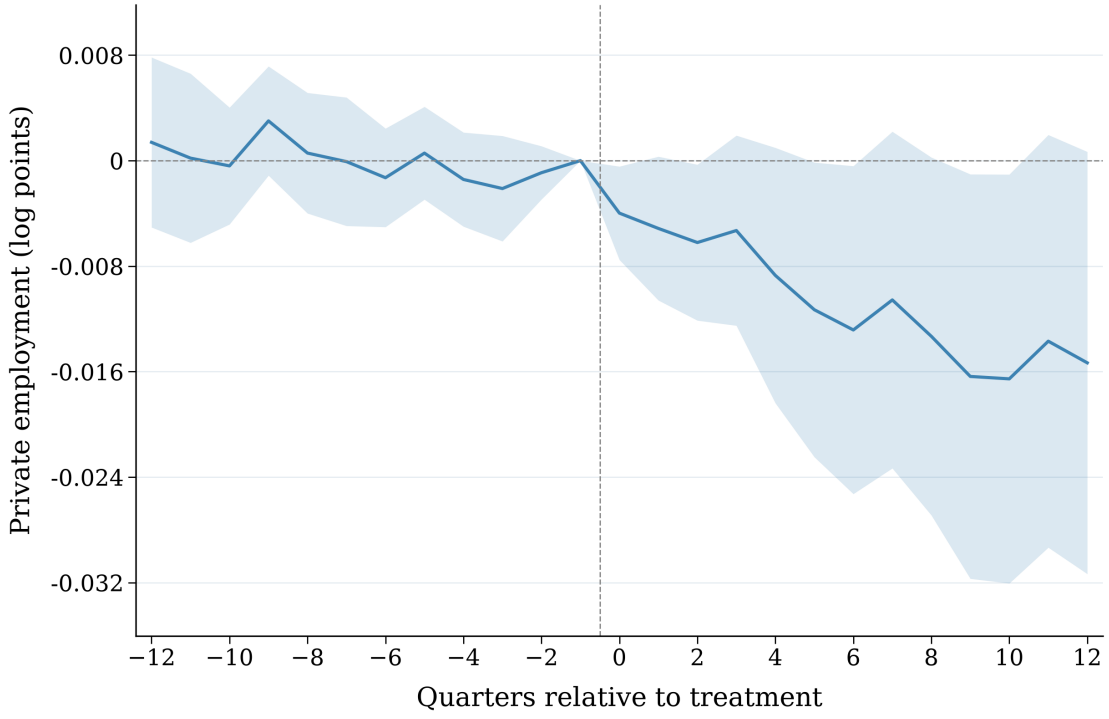


Notes: This figure presents dynamic effects on total expenditure (Panel a), current operations (Panel b), combined capital expenditure (Panel c), and services expenditure (Panel d). The figure plots the estimated coefficients (β_k) and 95% confidence intervals from the following stacked triple-difference regression estimated using PPML:

$$\mathbb{E}[Y_{isct} \mid X] = \exp\left[\sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \left(Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}\right) + \alpha_{ic} + \lambda_{stc} + \mu_{water_i \times t \times c}\right].$$

Y_{isct} represents the expenditure outcome in county i and year t in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed, and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption. Event year 0 is the year of adoption. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), state-by-year-by-cohort fixed effects (λ_{stc}), and wateradjacent-by-year-by-cohort fixed effects ($\mu_{water_i \times t \times c}$). The event window spans five years before and five years after the first post year. The data uses county-year observations from 1996-2022. The joint test that the coefficients for event years -5 through -2 equal zero is not rejected in any panel ($p = 0.633$, 0.734, 0.536, and 0.899 for Panels (a)–(d), respectively). The corresponding pooled estimates are reported in Table 4. Government expenditure data is collected from the Census Annual Survey of State and Local of State and Local Governments. Standard errors are clustered at the state level.

Figure 7. Event Study Estimates for Employment

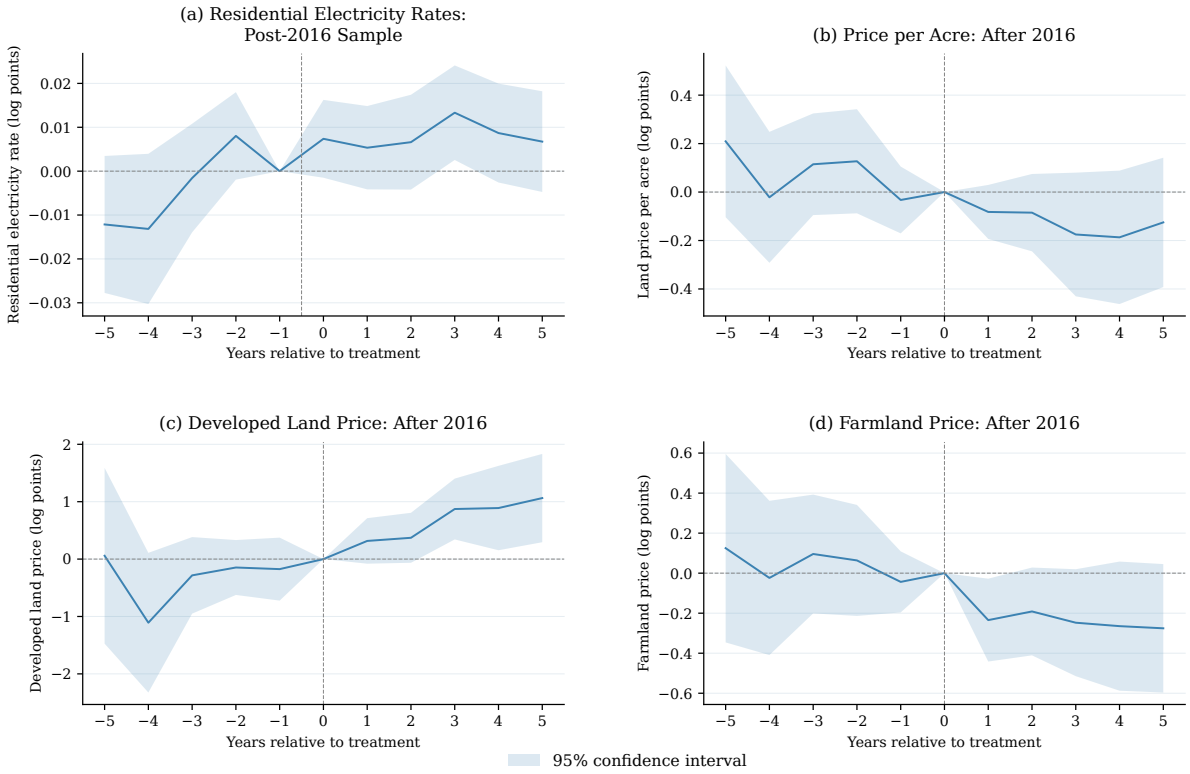


Notes: This figure presents dynamic effects on total employment. The figure plots the estimated coefficients (β_k) and shaded 95% confidence intervals from the following stacked triple-difference regression estimated using PPML:

$$\mathbb{E}[Y_{iscq} \mid \cdot] = \exp\left[\sum_{\substack{k=-12 \\ k\neq -1}}^{12} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{q - T_{sc} = k\}) + \alpha_{ic} + \lambda_{sqc} + \mu_{water_i \times q \times c}\right].$$

Y_{iscq} represents total employment in county i and quarter q in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed, and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{q - T_{sc} = k\}$ is an indicator for each event quarter k, where T_{sc} is the treatment quarter. Event quarter 0 is the first post-treatment quarter. Event quarter -1 is excluded as the reference quarter. All regressions include county-by-cohort fixed effects (α_{ic}), state-by-quarter-by-cohort fixed effects (λ_{sqc}), and water-by-quarter-by-cohort fixed effects ($\mu_{water_i \times q \times c}$). The event window spans 12 quarters before and 12 quarters after treatment. The data uses county-quarter observations through 2022. The corresponding pooled estimate is reported in Table 6, Panel A. Data is from the Quarterly Census of Employment and Wages. Standard errors are clustered at the state level.

Figure 8. Event Study Estimates for Residential Electricity Rates and Land Prices



Notes: This figure presents dynamic effects on log residential electricity rates (Panel a), log land prices per acre (Panel b), log developed land prices (Panel c), and log farmland prices (Panel d). The figure plots the estimated coefficients (β_k) and 95% confidence intervals from the following stacked triple-difference regression:

$$Y_{isct} = \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \cdot (Treat_{sc} \times WaterAdjacent_i \times \mathbf{1}\{t - T_{sc} = k\}) + \alpha_{ic} + \tau_{tc} + \lambda_{stc} + \mu_{water_i \times t \times c} + \varepsilon_{isct}.$$

Y_{isct} represents the log electricity rate or land price outcome in county i and year t in cohort stack c. $Treat_{sc}$ identifies counties in states where a data center subsidy was passed, and $WaterAdjacent_i$ identifies counties that share a border with a water body. The variable $\mathbf{1}\{t - T_{sc} = k\}$ is an indicator for each event year k, where T_{sc} is the calendar year of incentive adoption. Event year 0 is the year of adoption. Event year -1 is excluded as the reference year. The regression includes county-by-cohort fixed effects (α_{ic}), year-by-cohort fixed effects (τ_{tc}), state-by-year-by-cohort fixed effects (λ_{stc}), and wateradjacent-by-year-by-cohort fixed effects ($\mu_{water_i \times t \times c}$). The event window spans five years before and five years after the first post year. The data uses county-year observations from 2001-2024 for electricity rates and 2010-2024 for land prices. The corresponding pooled estimates are reported in Table 8 and Table 10. Data is from the Quarterly Census of Employment and Wages. Standard errors are clustered at the state level.

9 Tables

Table 1
Summary Statistics

Variable	Observations	Mean	Median	SD
Panel A: Data Center Investment				
Cumulative number of data centers (count)	90,740	0.267	0.000	2.815
Cumulative facility space (square feet)	90,740	31,050.04	0.00	470,815.40
Cumulative power capacity (MW)	90,740	5.358	0.000	107.835
Panel B: Government Revenue ($ thousands)				
Total revenue	73,172	278,161.28	61,085.00	757,060.98
Total taxes	73,172	113,083.17	18,371.00	331,455.37
General charges	73,172	46,193.54	6,947.00	138,861.24
Intergovernmental revenue	73,172	115,721.91	31,478.00	293,001.68
Sales taxes	67,637	14,320.18	843.00	49,323.34
Property taxes	73,195	86,907.59	14,605.00	254,223.20
Public utility taxes	56,429	2,695.36	144.00	10,963.43
License taxes	62,806	2,619.72	115.00	9,995.00
Utility revenue	66,233	27,615.67	2,466.00	101,300.93
Parks and recreation charges	59,938	1,773.34	80.00	6,071.05
Education charges	72,832	5,620.70	1,099.50	13,704.54
Sewerage charges	65,140	9,712.52	866.00	31,629.57
Panel C: Government Expenditure ($ thousands)				
Total expenditure	73,199	319,855.66	68,188.00	878,579.63
Direct expenditure	73,199	312,282.43	66,711.00	855,180.30
Current operations	73,199	257,101.12	57,312.00	694,016.29
Intergovernmental expenditure	70,991	7,251.82	736.00	25,018.82
Combined capital	73,153	40,631.19	6,449.00	122,819.18
Construction	71,189	29,898.00	3,843.00	92,628.60
Other capital outlays	71,707	9,716.24	2,063.00	25,446.78
Interest on debt	72,542	13,617.94	1,443.00	47,118.77
Utilities expenditure	66,448	33,761.47	2,738.50	133,648.15
Infrastructure expenditure	68,900	30,342.46	5,081.00	92,530.63
Services expenditure	73,188	205,108.03	49,096.50	533,179.83
Assistance and subsidies	18,071	2,564.56	1.00	10,875.48
Total education expenditure	71,575	138,558.50	38,590.00	328,144.29
Education construction	69,816	10,711.75	1,406.00	29,248.27
Panel D: Government Debt ($ thousands)				
Total long-term debt	72,482	305,503.38	33,420.00	1,044,415.14
Government-purpose long-term debt	72,467	256,372.59	29,593.00	885,909.25
Private-purpose long-term debt	36,087	90,070.23	4,416.00	273,173.36
Short-term debt	47,538	7,270.68	89.00	28,450.43

Table 1 (continued)

Variable	Observations	Mean	Median	SD
Panel E: Employment				
Quarterly establishments (count)	337,792	2,586.76	564.00	10,514.36
Average employment (workers)	337,792	34,812.66	6,274.17	128,854.60
Average wage ($ per quarter)	336,549	8,228.42	7,768.24	2,940.43
Panel F: Electricity Rates ($/kWh)				
Residential electricity rate	72,779	0.1113	0.1089	0.0349
Panel G: Land Prices ($ per acre)				
Overall land	27,512	37,607.65	7,667.88	406,026.43
Developed land	20,067	187,993.73	56,185.98	1,167,651.65
Farmland	27,012	20,052.16	7,862.03	393,427.24
Panel H: Municipal Bond Yield (percentage points)				
Volume-weighted yield	4,330,587	2.5157	2.4077	1.3790
Panel I: Education Outcomes (SD units)				
Reading scores	30,317	-0.055	-0.031	0.258
Math scores	29,887	-0.057	-0.038	0.293

Notes: This table reports the number of observations, mean, median, and standard deviation for the variables before stacking. Panel A uses county-year observations from 1996–2024 on data center investment from the Aterio U.S. inventory and is supplemented with information hand collected from Baxtel.com. Panels B–D use county-year observations from 1995–2022 on government revenue, expenditure, and year-end debt outstanding from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. The first four revenue measures use a common sample in which all four are reported. Other capital outlays and the two education expenditure measures are available through 2021. Revenue, expenditure, and debt are reported in thousands of nominal dollars. Panel E uses county-quarter observations from 1996–2022 on establishments, employment, and wages from the Bureau of Labor Statistics' Quarterly Census of Emloyment and Wages (QCEW). Panel F uses county-year observations from 2001–2024 on residential electricity rates from the U.S. Energy Information Administration (EIA), measured in dollars per kilowatt-hour. Panel G uses county-year observations from 2010–2025 on land prices from Acres.com, measured in dollars per acre. Panel H uses bond-month observations from 2005–2023 on municipal bond yields from the Municipal Securities Rulemaking Board (MSRB), with volume-weighted yields measured in percentage points. Panel I uses county-year observations from 2009–2025 on reading and math scores from the Stanford Education Data Archive (SEDA), excluding unavailable scores in 2020–2021. Scores are measured in standard-deviation units.

Table 2
Stacked Difference-in-Differences Estimates for Normalized Data Center Outcomes

	Normalized count	Normalized space	Normalized power
Post × Water-adjacent × Treated	0.303**	0.423*	0.502**
	(0.123)	(0.217)	(0.234)
County FE	Yes	Yes	Yes
State × year FE	Yes	Yes	Yes
Water-adjacent × year FE	Yes	Yes	Yes
Observations	251,279	251,279	251,279
Adjusted R^2	0.919	0.860	0.692

Notes: This table reports coefficients from annual stacked county-panel regressions of data center investment on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for years after incentive adoption (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. The dependent variables are cumulative data center count (Column 1), facility space in square feet (Column 2), and power capacity in megawatts (Column 3), each divided by its mean across unique county-years from 1996–2024. Each cohort's sample spans five years before through five years after the first calendar year following adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data uses county-year observations from 1996-2024. Facility data are from the Aterio U.S. inventory and supplemented with information hand collected from Baxtel.com. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$,and $^{***}p < 0.01$.

Table 3
Effects of Data Center Tax Incentives on Government Revenue

	(1)	(2)	(3)	(4)
Panel A: Revenue and Its Components	Total revenue	Total taxes	General charges	Intergov. revenue
Post × Water-adjacent × Treated	0.0203**	0.0248**	0.0434	0.0096
	(0.0100)	(0.0119)	(0.0304)	(0.0094)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.992	0.991	0.975	0.991
Observations	271,501	271,501	271,501	271,501
Panel B: Tax Revenue	Sales taxes	Property taxes	Public utility taxes	License taxes
Post × Water-adjacent × Treated	0.0540**	0.0190*	-0.0145	0.0079
	(0.0268)	(0.0115)	(0.0447)	(0.0367)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.979	0.992	0.975	0.967
Observations	248,917	271,621	193,314	235,525
Panel C: Utility Revenue and Service Charges	Utility revenue	Parks and recreation charges	Education charges	Sewerage charges
Post × Water-adjacent × Treated	0.0458**	0.0564	0.0058	0.0157
	(0.0219)	(0.0393)	(0.0139)	(0.0265)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.981	0.968	0.987	0.976
Observations	231,185	208,895	269,882	226,695

Notes: This table reports coefficients from annual stacked county-panel PPML regressions of government revenue on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the adoption year and subsequent years (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-by-year fixed effects. All fixed effects are interacted with cohort indicators. Total revenue equals taxes, general charges (including liquor), and intergovernmental revenue. Panel A uses a common sample in which total revenue, total taxes, general charges, and intergovernmental revenue are all reported. Sales taxes comprise general and other selective sales taxes. Each cohort's sample spans five years before and five years after adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data uses county-year observations from 1995-2022. Revenue data are from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4
Effects of Data Center Tax Incentives on Expenditure

	(1)	(2)	(3)	(4)
Panel A: Overall expenditure	Total expenditure	Direct expenditure	Current operations	Intergov. expenditure
Post × Treated × Water-adjacent	0.0308**	0.0298**	0.0206**	-0.0350
	(0.0123)	(0.0123)	(0.0099)	(0.0360)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	271,637	271,637	271,637	256,868
Pseudo R^2	0.992	0.992	0.993	0.961
Panel B: Capital and financing	Combined capital	Construction	Other capital outlays	Interest on debt
Post × Treated × Water-adjacent	0.0549**	0.0539*	0.0847**	-0.0100
	(0.0262)	(0.0314)	(0.0368)	(0.0348)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	271,406	260,723	263,682	267,635
Pseudo R^2	0.964	0.951	0.943	0.983
Panel C: Services and assistance	Utilities expenditure	Infrastructure expenditure	Services expenditure	Assistance and subsidies
Post × Treated × Water-adjacent	0.0362	0.0128	0.0226**	0.1955***
	(0.0322)	(0.0268)	(0.0106)	(0.0742)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	232,664	247,834	271,593	42,024
Pseudo R^2	0.980	0.973	0.992	0.937
Panel D: Education expenditure and learning outcomes	Total education expenditure	Education construction	Reading scores	Math scores
Post × Treated × Water-adjacent	0.0176**	0.0786*	0.0040	0.0118**
	(0.0089)	(0.0448)	(0.0044)	(0.0057)
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes
Observations	263,050	252,633	172,496	173,022
Pseudo R^2	0.992	0.880	–	–
Adjusted R^2	–	–	0.894	0.888

Notes: This table reports coefficients from annual stacked county-panel PPML regressions of government expenditure and OLS regressions of education outcomes on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the adoption year and subsequent years (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects, all interacted with cohort indicators. Total expenditure includes intergovernmental transfers, while direct expenditure excludes them. Each cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Reading and math scores cover 2009–2025 and are from the Stanford Education Data Archive (SEDA). Comparison states are never treated or not yet treated at cohort adoption. Expenditure data covers 1995-2022 and is from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 5
Effects of Data Center Tax Incentives on Government Debt and Municipal Borrowing Costs

	Debt outstanding				Municipal borrowing costs	
	Long-term: total	Long-term: government purpose	Long-term: private purpose	Short-term	Bond yield	Treasury-adjusted yield spread
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Treated × Water-adjacent	0.0097	0.0535***	-0.1114	-0.0798	0.110**	0.104*
	(0.0234)	(0.0187)	(0.0726)	(0.1423)	(0.050)	(0.053)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.986	0.987	0.952	0.937	—	—
Adjusted R^2	—	—	—	—	0.382	0.240
Observations	267,302	267,226	111,152	141,825	46,878,051	46,878,051

Notes: This table reports coefficients from annual stacked county-panel PPML regressions of government debt and monthly OLS regressions of municipal bond yields and yield spreads on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the post-adoption period (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-time, and water-adjacent-by-time fixed effects, all interacted with cohort indicators. Time denotes years for the debt regressions and months for the municipal bond regressions. Debt measures year-end outstanding balances. Municipal bond yields and Treasury-adjusted yield spreads are measured in percentage points. Each debt cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data uses county-year observations from 1995-2022. Revenue data are from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Municipal bond data are from the Municipal Securities Rulemaking Board (MSRB). Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 6
Effects of Data Center Tax Incentives on Employment, Establishments, and Wages

	Employment	Establishments	Average quarterly wages
	(1)	(2)	(3)
Panel A: All Industries			
Post × Treated × Water-adjacent	-0.0105*	-0.0075**	-0.0053**
	(0.0063)	(0.0037)	(0.0025)
Pseudo R^2	0.9996	0.9985	0.9500
Observations	584,169	585,819	585,819
Panel B: Electric Power			
Post × Treated × Water-adjacent	0.0836*	-0.0108	0.0193*
	(0.0470)	(0.0325)	(0.0117)
Pseudo R^2	0.9809	0.5722	0.8586
Observations	55,240	55,240	55,240
Panel C: Natural Gas Distribution			
Post × Treated × Water-adjacent	0.6217***	0.0609	0.2059***
	(0.2275)	(0.0842)	(0.0798)
Pseudo R^2	0.9729	0.6575	0.8426
Observations	9,144	9,144	9,144
Panel D: HVAC and Semiconductors			
Post × Treated × Water-adjacent	0.1048*	0.0732**	-0.0197
	(0.0602)	(0.0314)	(0.0242)
Pseudo R^2	0.9886	0.7941	0.8657
Observations	48,770	48,770	48,770
Panel E: Residential Construction			
Post × Treated × Water-adjacent	-0.0389***	-0.0208**	0.0028
	(0.0131)	(0.0093)	(0.0101)
Pseudo R^2	0.9845	0.9641	0.8549
Observations	344,819	344,819	344,819
County FE	Yes	Yes	Yes
State × Quarter FE	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes

Notes: This table reports coefficients from quarterly stacked county-panel PPML regressions of employment, establishments, and wages on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the post-adoption period (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-quarter, and water-adjacent-by-time fixed effects, all interacted with cohort indicators. Each cohort's sample spans 12 quarters before through 12 quarters after the adoption of data center incentive, with Post equal to one beginning in that quarter. Comparison states are never treated or not yet treated at cohort adoption. The data use county-quarter observations through 2022 and are from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 7
Effects of Data Center Tax Incentives on Employment, Establishments, and Wages by Industry Group

	Employment (1)	Establishments (2)	Wages (3)
Panel A: Complementary industries			
Post × Treated × Water-adjacent	0.0593***	0.0169	-0.0047
	(0.0223)	(0.0173)	(0.0106)
Observations	407,688	407,688	407,688
Pseudo R^2	0.894	0.768	0.780
Panel B: Resource-intensive industries			
Post × Treated × Water-adjacent	-0.0042	-0.0103***	-0.0034*
	(0.0101)	(0.0038)	(0.0020)
Observations	8,409,364	8,409,364	8,409,364
Pseudo R^2	0.889	0.911	0.778
Panel C: Other industries			
Post × Treated × Water-adjacent	-0.0098	-0.0046	-0.0042*
	(0.0065)	(0.0052)	(0.0024)
Observations	25,281,114	25,281,114	25,281,114
Pseudo R^2	0.897	0.900	0.740
County × Cohort FE	Yes	Yes	Yes
State × Quarter × Cohort FE	Yes	Yes	Yes
Water-adjacent × Quarter × Cohort FE	Yes	Yes	Yes
Industry × State × Year × Cohort FE	Yes	Yes	Yes

Notes: This table reports coefficients from stacked county–industry–quarter PPML regressions of employment, establishments, and wages on the interaction between an indicator for whether the county is in a state belonging to the adoption cohort (Treated), an indicator for the post-adoption period (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). Panel A includes HVAC equipment, semiconductors and other electronic components, electric power, natural gas, water and sewage systems, and telecommunications. Panel B includes resource-intensive industries and Panel C includes the remaining eligible industries. Each cohort's sample spans 12 quarters before through 12 quarters after that start. The sample includes observations through 2022. Data are from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages. Standard errors clustered at the state level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 8
Effect of Data Center Exposure on Electricity Rates and Land Prices

	Residential electricity	Developed land	Agricultural land
Post × Water-adjacent × Treated	0.002	0.198**	-0.008
	(0.004)	(0.045)	(0.037)
County FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes
Observations	245,883	44,198	62,359
Adjusted R^2	0.913	0.574	0.715

Notes: This table reports coefficients from annual stacked county-panel OLS regressions of log residential electricity rates, log developed-land prices, and log agricultural-land prices on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the post-adoption period (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. Each cohort's sample spans five years before through five years after adoption. Comparison states are never treated or not yet treated at cohort adoption. The table reports full-sample estimates. Electricity rate data use county-year observations from 2001–2024 and are from the U.S. Energy Information Administration (EIA). Land price data use county-year observations from 2010–2025 and are from Acres.com. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 9
Data Center Tax Incentives, Rents, and Home Values

	Log median gross rent (1)	Log median home value (2)
Post × Treated × Water-adjacent	-0.0079**	-0.0095*
	(0.0038)	(0.0053)
County FE	Yes	Yes
State × Year FE	Yes	Yes
Water-adjacent × Year FE	Yes	Yes
Observations	205,802	205,734
Adjusted R^2	0.9493	0.9831

Notes: This table reports coefficients from annual stacked county-panel OLS regressions of log median monthly gross rent and log median owner-occupied home value on the interaction between an indicator for whether the county is in a state that has adopted a data center incentive (Treated), an indicator for the adoption year and subsequent years (Post), and an indicator for whether the county is water-adjacent (Water-adjacent). The regression also includes county, state-by-year, and water-adjacent-by-year fixed effects. All fixed effects are interacted with cohort indicators. Each cohort's sample spans five years before through five years after adoption, with Post equal to one beginning in the year of treatment. Comparison states are never treated or not yet treated at cohort adoption. The data use county-year observations from the American Community Survey five-year releases from 2009–2022. Dollar amounts are nominal. Standard errors clustered at the state level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 10
Effect of Data Center Exposure on Electricity Rates, Land Prices, Employment, and Political Outcomes by Period

	Residential electricity	Developed land	Agricultural land	Employment	Incumbent lost
Panel A: Pre-2016					
Post × Water-adjacent × Treated	-0.002	0.473*	-0.004	0.001	-0.042**
	(0.006)	(0.253)	(0.094)	(0.004)	(0.018)
County FE	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes	Yes
Observations	175,921	12,504	29,942	275,051	85,605
Adjusted R^2 / Pseudo R^2	0.890	0.611	0.750	0.9996	0.052
Panel B: Post-2016					
Post × Water-adjacent × Treated	0.008*	0.355**	-0.213**	-0.011*	0.076***
	(0.004)	(0.175)	(0.100)	(0.007)	(0.028)
County FE	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Year FE	Yes	Yes	Yes	Yes	Yes
Observations	68,419	20,863	31,716	241,860	24,740
Adjusted R^2 / Pseudo R^2	0.931	0.597	0.676	0.9997	0.078

Notes: This table reports estimates separately for observations before 2016 and from 2016 onward. The outcomes are the log residential electricity rate, log developed-land price, log agricultural-land price, the PPML regression of Employment, and an indicator for whether the incumbent lost. Standard errors clustered by state are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 11
Robustness of Data Center, Fiscal, and Employment Estimates

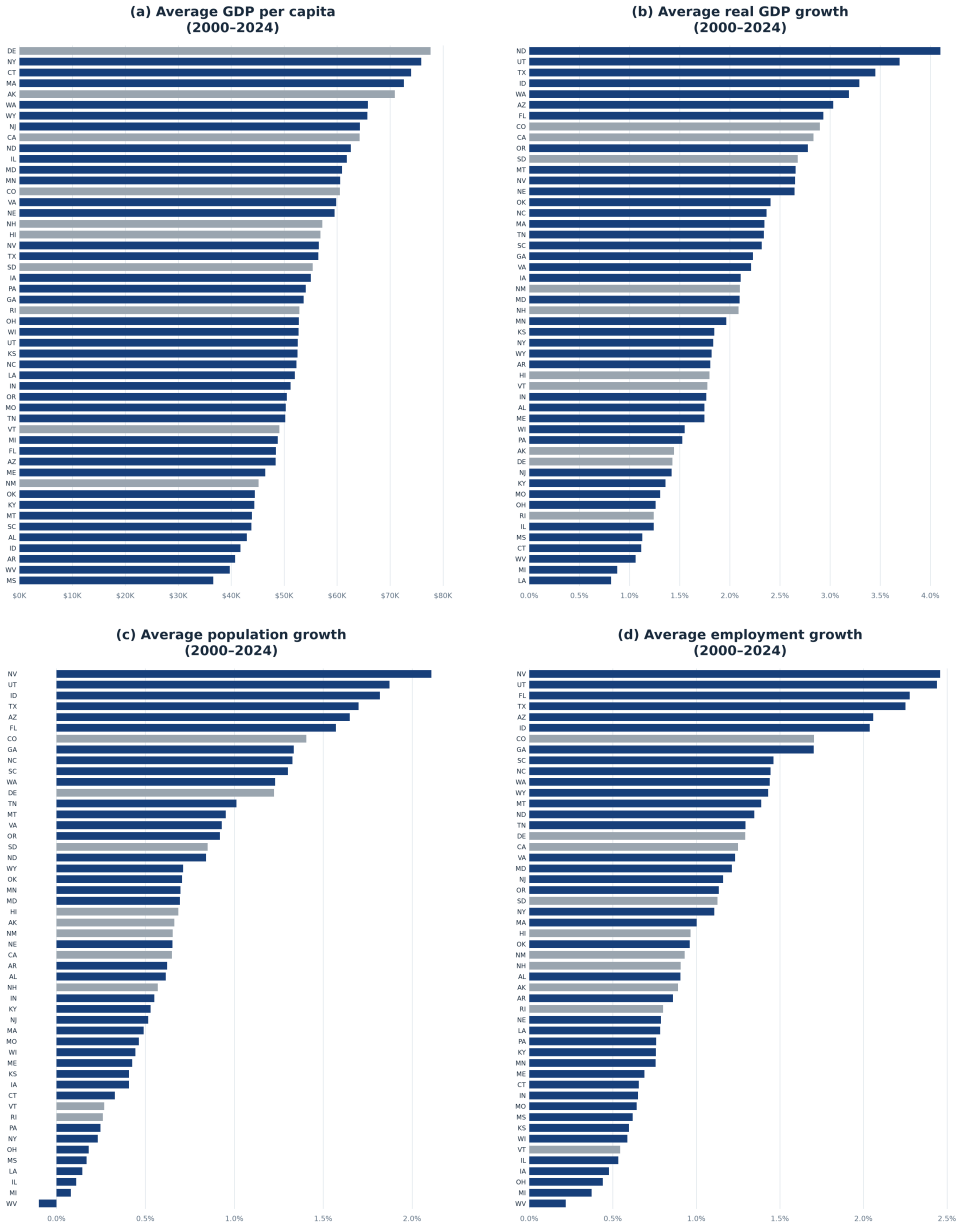
	(1)	(2)	(3)	(4)	(5)
Panel A: Excluding states with evidence of lobbying	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Water-adjacent	0.2043	0.0240*	0.0334**	0.0334*	-0.0032
	(0.1218)	(0.0137)	(0.0162)	(0.0180)	(0.0040)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.934	0.993	0.991	0.992	0.9996
Observations	170,416	180,335	180,335	180,406	382,713
Panel B: Excluding states with incentives enacted in broader legislative packages	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Water-adjacent	0.3081**	0.0237*	0.0233	0.0459***	-0.0127
	(0.1407)	(0.0130)	(0.0174)	(0.0178)	(0.0084)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.891	0.990	0.988	0.990	0.9995
Observations	176,591	188,653	188,653	188,758	393,395
Panel C: Dillon's Rule states	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Water-adjacent	0.5122*	0.0351**	0.0484***	0.0515***	-0.0206***
	(0.2497)	(0.0163)	(0.0166)	(0.0174)	(0.0068)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Water-adjacent × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.914	0.992	0.992	0.992	0.9996
Observations	59,025	63,592	63,592	63,652	133,080

Notes: Each panel reports separate stacked triple-difference regressions. Column 1 uses annual OLS through 2024; the data center count is divided by its 1996–2024 county–year mean (0.26691), and event year zero is the year of adoption. Columns 2–4 use annual PPML through 2022, with adoption as event year zero. Columns 2–3 require total revenue, total taxes, general charges, and intergovernmental revenue to be reported. Column 5 uses quarterly PPML for average private employment. Time fixed effects refer to years in Columns 1–4 and quarters in Column 5; all fixed effects are interacted with cohort. All robustness samples exclude California. Standard errors are clustered by state. Panel A excludes VA, TX, OH, GA, IL, IN, MI, NC, WI, SC, and CT. Panel B excludes FL, IL, KY, MA, MN, NJ, NY, NC, OH, OR, PA, SC, and WI. Panel C restricts the sample to AZ, DE, ID, NE, NH, NJ, NC, OK, SD, TX, VT, VA, WV, and WY, using NACo's county-authority classifications; Utah is excluded because its Hutchinson's Rule grants broader local authority. The classification is fixed rather than reconstructed annually. Parentheses report standard errors. Fit statistics are adjusted R^2 for OLS and pseudo R^2 for PPML. Observations count stacked county–years in Columns 1–4 and county–quarters in Column 5. $^{*}p < 0.10$, $^{**}p < 0.05$,and $^{***}p < 0.01$.

the

A Additional Figures

Figure A.1. Distribution of Data Center Incentive States.



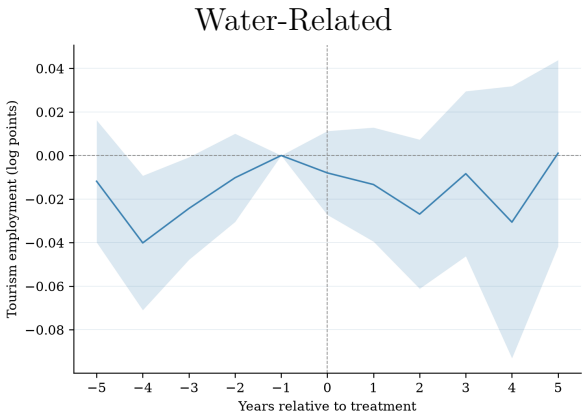
*Notes:*This figure ranks states according to their average economic characteristics between 2000 and 2024. Panels (a) through (d) present average GDP per capita, real GDP growth, population growth, and employment growth, respectively. Dark-blue bars represent states that adopted a data center tax incentive, while gray bars represent states that have not. GDP per capita is reported in thousands of 2017 dollars, and growth rates are reported in percentage terms. The District of Columbia is excluded from the sample.

Figure A.2. State-Level Economic Trends

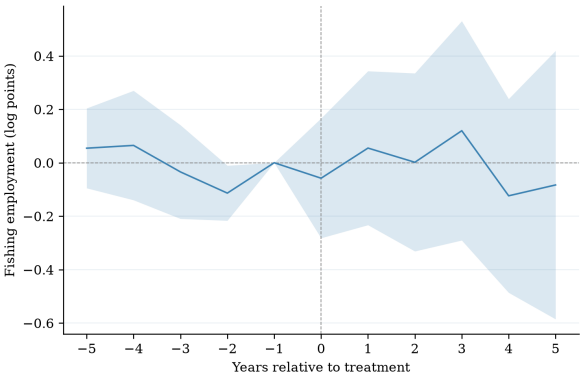


Notes: This figure compares economic trends between 2000 and 2024 in states that adopted data center tax incentives with trends in states that never adopted a data center tax incentive. Panels (a) through (d) present average GDP per capita, real GDP growth, population growth, and employment growth, respectively. Dark-blue lines represent incentive states, while gray lines represent never treated states. Shaded areas report 95% confidence intervals calculated from cross-state variation. The District of Columbia is excluded from the sample.

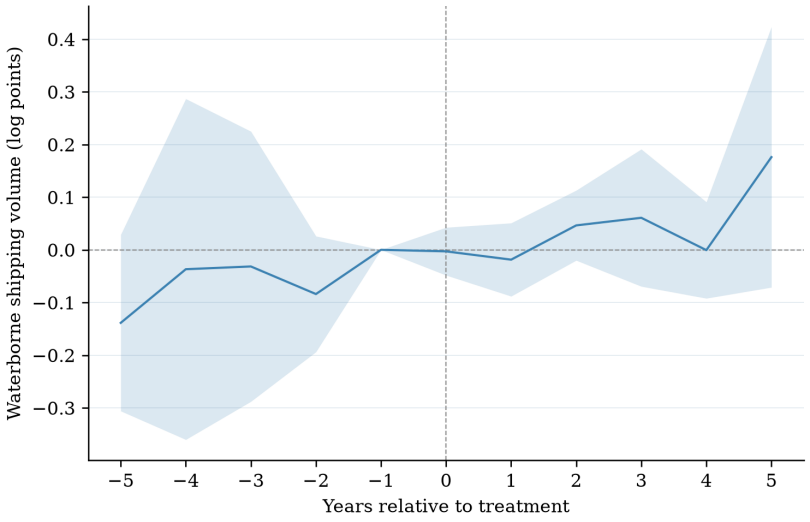
Figure A.3. Placebo Tests for Wate Related Economic Activity



(a) Tourism employment



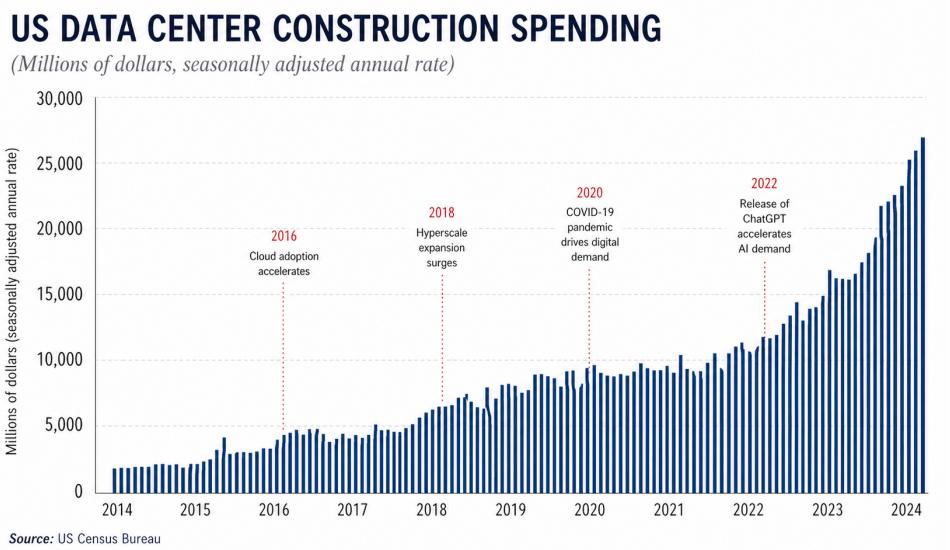
(b) Fishing employment



(c) State waterborne shipping volume

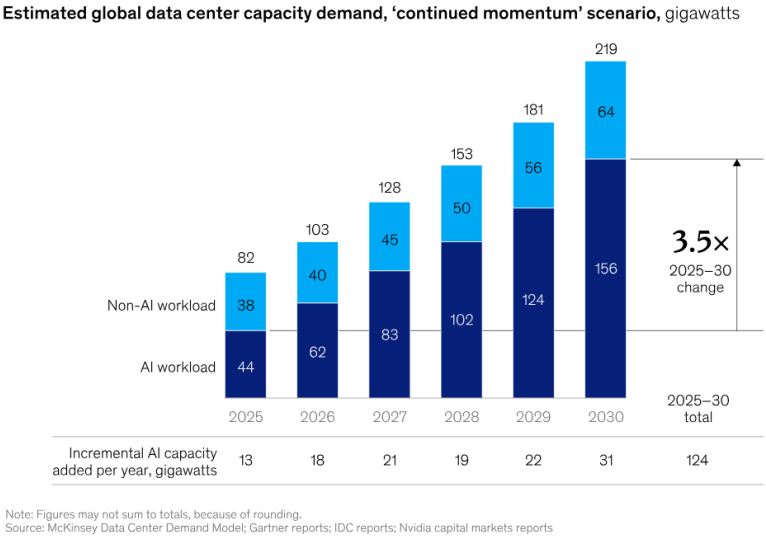
Notes: This Figure reports coefficients from a stacked county-panel regression of water-related economic activity on the interaction between an indicator for whether the state passed a data center subsidy(Treated) and an indicator for years equal to or after the passage of the subsidy(Post) and an indicator for whether the county is adjacent to a water body (Water-Adjacent), with county-by-cohort and year-by-cohort fixed effects. The sample includes county-year observations from 1996-2024. Standard errors are clustered at the state level. All outcomes are expressed in natural logarithms. Event year −1 is omitted in every panel. Shaded areas report 95 percent confidence intervals, and standard errors are clustered by state.

Figure A.4. Data Center Construction Spending



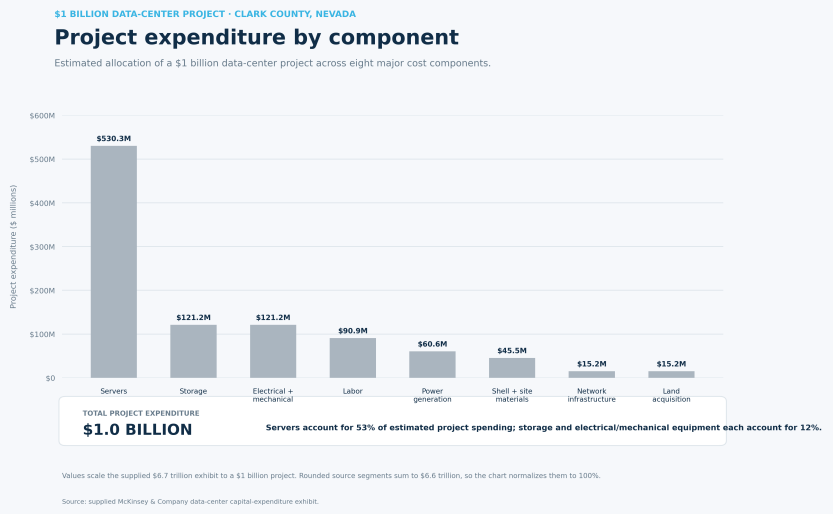
Notes: This figure reports U.S. data center construction spending from 2014 through 2024, measured in millions of dollars at a seasonally adjusted annual rate. The annotations highlight developments in cloud adoption in 2016, hyperscale expansion in 2018, digital demand during the COVID-19 pandemic in 2020, and the release of ChatGPT in 2022. Data is from the U.S. Census Bureau.

Figure A.5. Data Center Projected Growth

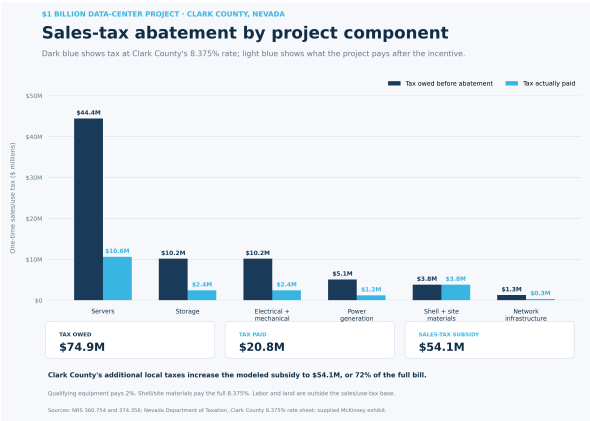


Notes: This figure reports projected global demand for data center capacity from 2025 through 2030 under McKinsey's "continued momentum" scenario. Capacity is measured in gigawatts, with dark blue indicating AI workloads and light blue indicating non-AI workloads. Total capacity demand is projected to increase from 82 gigawatts in 2025 to 219 gigawatts in 2030, while capacity demand for AI workloads increases from 44 to 156 gigawatts. The figures below the bars report incremental AI capacity added each year. Projections are from the McKinsey Data Center Demand Model.

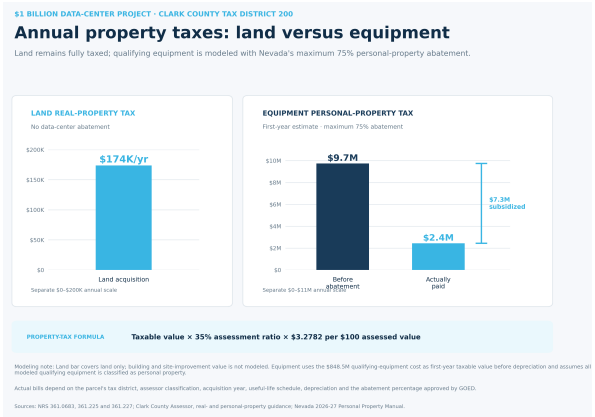
Figure A.6. Example of a Data Center Tax Incentive in Clark County, Nevada.



(a) Project expenditures by component



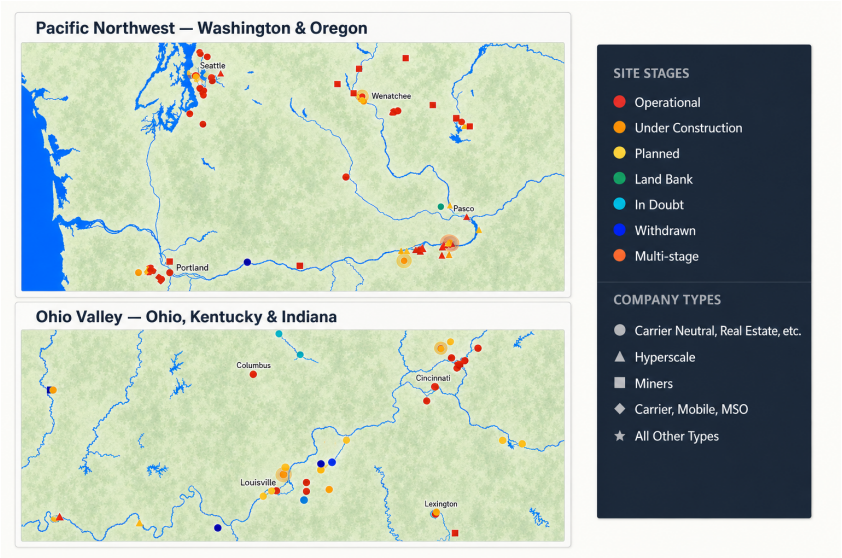
(b) Sales and use tax by component



(c) Real and personal property taxes

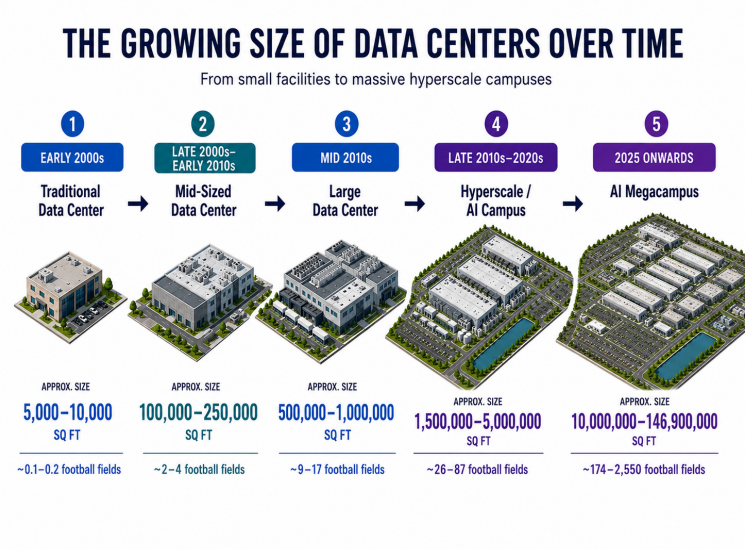
Notes: This figure illustrates the tax treatment of a hypothetical data center project in Clark County, Nevada. Panel (a) presents project expenditures by component. Panel (b) compares the sales and use taxes owed under the standard tax rate with the amount paid after the abatement. Panel (c) presents the corresponding real and personal property-tax liabilities.

Figure A.7. Data center locations and proximity to water bodies



Notes: This figure shows data center locations and nearby water bodies in the Pacific Northwest, covering Washington and Oregon (upper panel), and the Ohio Valley, covering Ohio, Kentucky, and Indiana (lower panel). Marker colors indicate project stages, while marker shapes distinguish company types, as shown in the legend. Water bodies are shown in blue.

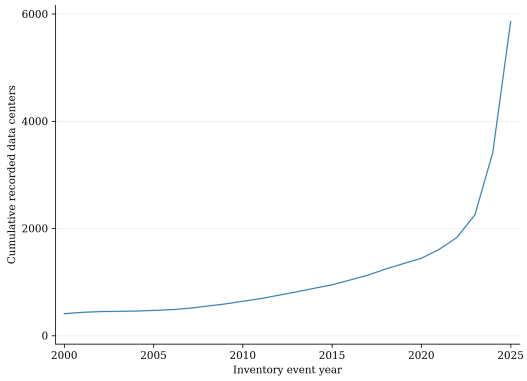
Figure A.8. Data Centers Over Time



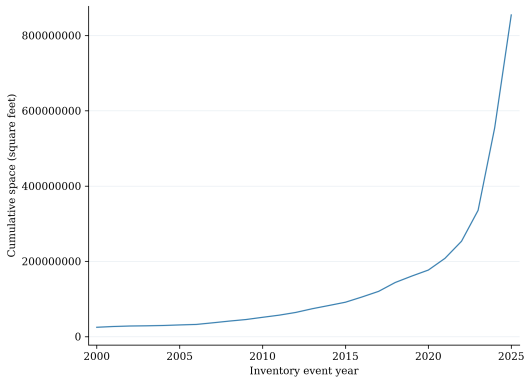
Notes: This figure illustrates the evolution of data center scale from traditional facilities in the early 2000s to hyperscale and artificial intelligence campuses. The timeline shows facility and campus types and approximate sizes reported in square feet and expressed as equivalent numbers of football fields.

Figure A.9. Growth in Data Center Scale Over Time

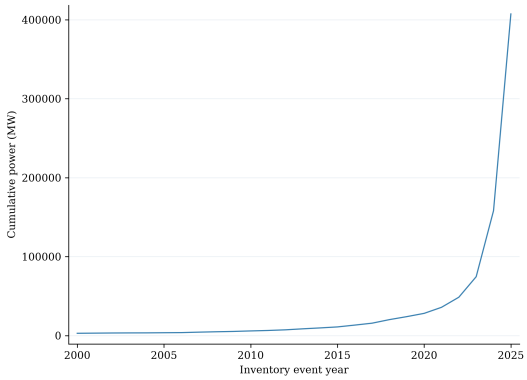
(a) Cumulative Recorded Data Centers



(b) Total Data Center Space

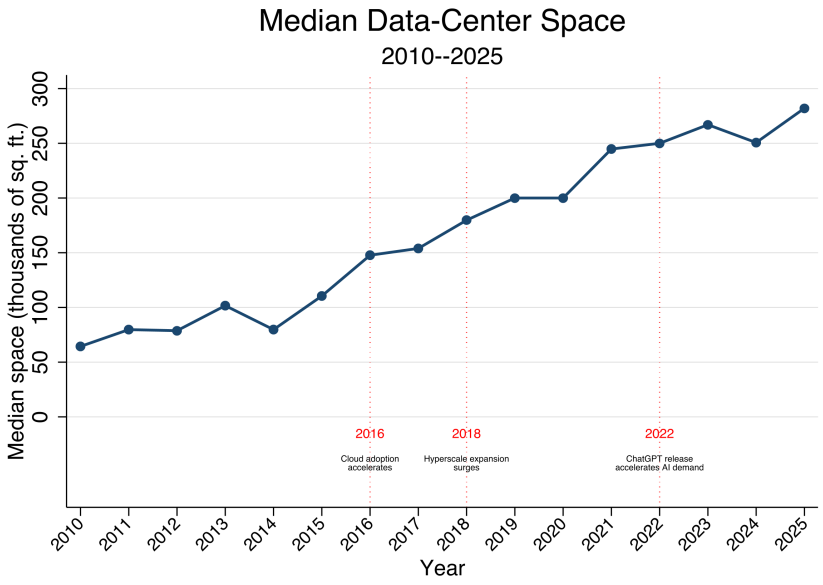


(c) Estimated Power Capacity

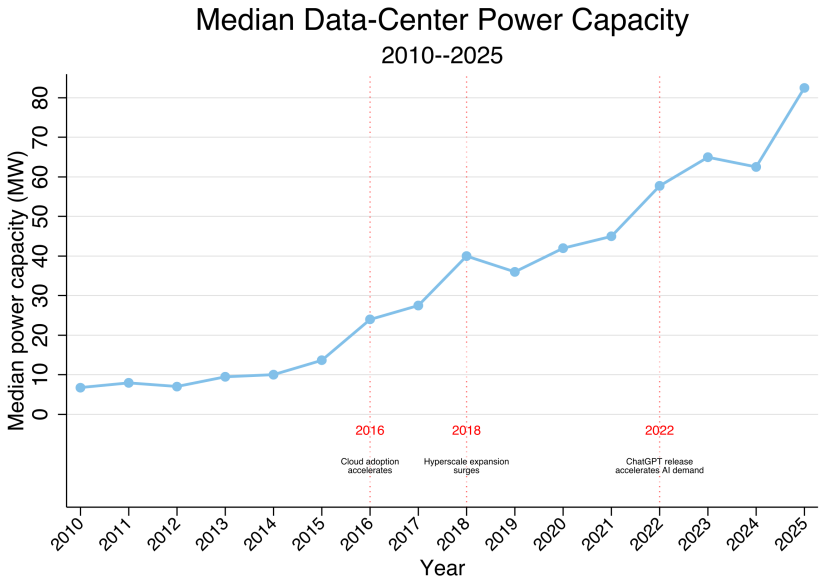


Notes: This figure plots the growth of data center activity from 2000 to 2025. Panel A reports cumulative recorded data center development events, Panel B reports cumulative data center space in square feet, and Panel C reports estimated cumulative power capacity in megawatts. The figures show that recent data center growth reflects not only an increase in the number of facilities, but also a sharp increase in facility scale and power intensity.

Figure A.10. New Data Center Scale and Power Capacity



(a) Median Data Center Space



(b) Median Data Center Power Capacity

Notes: This figure presents annual median data center size and power capacity of new data centers from 2010 through 2025. Panel (a) reports median data center space in thousands of square feet. Panel (b) reports median selected power capacity in megawatts. Red dotted lines identify important periods in the evolution of the data center industry: the acceleration of cloud adoption in 2016, the expansion of hyperscale data centers in 2018, and the release of ChatGPT and subsequent acceleration of AI-related demand in 2022. Annual medians are calculated across observations with nonmissing values.

B Additional Tables

Table B.1
Description of Key Variables

This table reports the definitions and sources of the key variables used in the analysis.

Variable	Description	Source
Number of Data Centers	Cumulative number of recorded data center projects in county c by year t. A project enters in its announcement year, or, if unavailable, construction-start, construction-finish, then activation year, in that order. Counts include announced and under-construction facilities, not only operating facilities.	Aterio; Baxtel.com
Data Center Space	Cumulative recorded facility square footage in the county, using the same project dates as the count measure.	Aterio; Baxtel.com
Data Center Power Capacity	Cumulative selected power capacity in megawatts (MW), using the same project dates as the count measure.	Aterio; Baxtel.com
Normalized Data Center Outcomes	Each cumulative outcome is divided by its mean over unique county–years in 1996–2024: 0.26691 facilities, 31,050.042 square feet, or 5.358 MW. The denominators include zero outcomes and are fixed across robustness checks. The space regression excludes observations with positive counts but zero space.	Author calculations
Treated State	Within each stacked cohort, an indicator for states belonging to the focal incentive-adoption cohort.	State statutes
Post	Equals one in the adoption year and subsequent years; event year 0 is adoption.	
Water-adjacent County	Indicator for the Water-adjacent county classification used in the regressions.	Geographic data; author classification
Water-adjacent Neighbor	Indicator for a county that borders a water-adjacent county but is not itself classified as water-adjacent.	U.S. Census Bureau
Population	Annual estimate of the total resident population of county c in year t.	U.S. Census Bureau
Revenue Measurement	Financial outcomes are annual county-level aggregates in thousands of nominal dollars. The revenue table uses available observations for each outcome in 1995–2022. Reported zeros are retained; general charges and property taxes are filled with zero when missing and total revenue is reported, and missing sales-tax components are filled with zero when total taxes are reported.	U.S. Census Bureau; author calculations

Continued on next page

Table B.1 continued

Variable	Description	Source
Total Revenue	Constructed sum of total taxes, general charges including liquor-store revenue, and intergovernmental revenue. This measure excludes separately recorded utility revenue and miscellaneous general revenue; it is not the broader Census revenue aggregate.	U.S. Census Bureau
Total Taxes	All reported tax receipts (Census item codes T*), including property, sales, income, license, and other taxes.	U.S. Census Bureau
General Charges	Receipts from government services, plus liquor-store revenue under the study definition. Public utility receipts are recorded separately. This differs from the official general-charges aggregate because liquor-store revenue is included.	U.S. Census Bureau
Intergovernmental Revenue	Transfers received from federal, state, and other local governments (B*, C*, and D*).	U.S. Census Bureau
Sales Taxes	General sales taxes plus other selective sales taxes (T09 + T19). This is narrower than all sales and gross-receipts taxes and does not include every selective sales-tax category.	U.S. Census Bureau
Property Taxes	Property-tax receipts (T01).	U.S. Census Bureau
Public Utility Taxes	Tax receipts classified as public utility taxes (T15). These differ from receipts earned by publicly operated utilities and from utility-license taxes.	U.S. Census Bureau
License Taxes	License-tax receipts across reported license categories (T20, T21, T22, T23, T24, T25, T27, T28, and T29).	U.S. Census Bureau
Utility Revenue	Receipts from publicly operated water, electricity, gas, and transit systems (A91–A94), excluding intergovernmental transfers.	U.S. Census Bureau
Parks and Recreation Charges	Charges for parks and recreation services (A61), excluding intergovernmental transfers. These are service receipts, not all funding allocated to parks.	U.S. Census Bureau
Education Charges	Identified education-service charges (A09, A10, A12, A16, A18, and A21), excluding intergovernmental transfers and general tax funding.	U.S. Census Bureau
Sewerage Charges	Charges for sewerage services (A80), excluding intergovernmental transfers.	U.S. Census Bureau
Total Expenditure	Direct expenditure plus intergovernmental expenditure, excluding insurance-trust benefits. Includes current operations, combined capital outlays, interest, assistance and subsidies, and transfers to other governments.	U.S. Census Bureau
Direct Expenditure	Expenditure excluding intergovernmental transfers and insurance-trust benefits. Interest is included once.	U.S. Census Bureau
Current Operations	Routine operating expenditure, including employee compensation, supplies, and purchased services.	U.S. Census Bureau
Combined Capital Expenditure	Construction plus other capital outlays. Historical capital totals are retained through 2011; F and G items are combined in 2012–2021; F items represent combined capital from 2022.	U.S. Census Bureau

Continued on next page

Table B.1 continued

Variable	Description	Source
,Construction Expenditure	Separately reported construction expenditure through 2021. In the overall construction regression, missing amounts are set to zero when current operations are observed and year is at least 2021; this is an analytical imputation, not separately observed construction after 2021.	U.S. Census Bureau
Other Capital Outlays	Capital expenditure excluding construction, including equipment and land purchases. Historical capital totals less construction through 2011; G items in 2012–2021. Not separately available from 2022.	U.S. Census Bureau
Interest on Debt	Interest payments on debt. Excludes principal repayments and debt outstanding.	U.S. Census Bureau
Assistance and Subsidies	Reported assistance and subsidy expenditure in harmonized J items. Coverage is more limited than for aggregate expenditure.	U.S. Census Bureau
Intergovernmental Expenditure	Transfers paid to other governments. Adding transfers to direct expenditure gives gross expenditure, not a consolidated total net of transfers across governments.	U.S. Census Bureau
Utilities Expenditure	Direct spending on publicly operated water, electricity, gas, and transit systems. Excludes transfers.	U.S. Census Bureau
Infrastructure Expenditure	Direct spending on highways, airports, ports, parking, sewerage, and solid waste. Utilities are separate. Excludes transfers.	U.S. Census Bureau
Services Expenditure	Direct spending on police, fire, parks, libraries, education, health, hospitals, welfare, and housing/community development. Excludes transfers.	U.S. Census Bureau
Total Education Expenditure	Education expenditure including current operations, capital expenditure, applicable other direct spending, and intergovernmental transfers.	U.S. Census Bureau
Education Construction Expenditure	Separately reported construction spending on education facilities, available through 2021. Missing values are not filled using the overall construction rule.	U.S. Census Bureau
Reading Scores	All-student reading achievement in cohort-scale standard-deviation units. District scores are aggregated to county–years using subject-specific assessment counts, among geographically matched single-county districts. Available years are 2009–2025, excluding 2020–2021.	Stanford Education Data Archive (SEDA 2025.2)
Math Scores	All-student math achievement in cohort-scale standard-deviation units. District scores are aggregated to county–years using subject-specific assessment counts, among geographically matched single-county districts. Available years are 2009–2025, excluding 2020–2021.	Stanford Education Data Archive (SEDA 2025.2)
Average Yield	Volume-weighted average yield for a CUSIP in a given month. Volume refers to the par value of the trade.	MSRB

Continued on next page

Table B.1 continued

Variable	Description	Source
Yield Spread	Calculated as the difference between the *Average Yield* and the coupon-equivalent risk-free yield. The risk-free yield is calculated using the present value of the bond's coupon payments and face value, discounted using the U.S. Treasury yield curve and maturity-matched zero-coupon yields from Gürkaynak et al. (2007). The yield spread is calculated following Longstaff et al. (2005): $\text{Yield Spread}_{i,t} = \text{Average Yield}_{i,t} - \text{Risk-Free Yield}_{i,t}$.	MSRB, FEDS

Table B.2
State Data Center Tax Incentives

State	Year	State Sales	Local Sales	Local Property Taxes	Investment Threshold	Jobs	Wage Requirement
AL	2012	Yes	Yes	Yes	$200M; $400M	20	$40K+
AZ	2013	Yes	Yes	No	$25M; $50M	None	None
AR	2023	Yes	Yes	No	$500M	25	None
CT	2021	Yes	None	No	$200M	30	None
FL	2017	Yes	Yes	No	$150M	20	None
GA	2018	Yes	Yes	No	$250M	25	None
ID	2020	Yes	None	No	$250M	30	None
IL	2019	Yes	Yes	No	$250M	20	120% of median wage
IN	2019	Yes	None	No	$25M; $150M	None	None
IA	2009	Yes	Yes	No	$200M; $1B	10; 50	None
KS	2025	Yes	Yes	No	$250M	20	None
KY	2023	Yes	None	No	$100M; $450M	None	None
LA	2024	Yes	Yes	Yes	$200M	50	None
ME	2021	Yes	None	No	None	20	None
MD	2020	Yes	None	No	$2M	5	None
MA	2024	Yes	None	No	$50M	100	None
MI	2016	Yes	None	No	$250M	30	150% of median wage
MN	2012	Yes	None	Yes	$30M	None	None
MS	2010	Yes	No	No	$20M	20	150% of mean wage
MO	2015	Yes	Yes	No	$25M	10	None
MT	2017	None	None	Yes	$50M	None	None
NE	2012	Yes	Yes	Yes	None	None	None
NV	2015	Yes	Yes	Yes	$25M; $100M	10; 50	None
NJ	2024	Yes	None	No	None	None	None
NY	2000	Yes	Yes	No	None	None	None
NC	2016	Yes	Yes	No	$75M	None	None
ND	2021	Yes	Yes	No	$150M	None	None
OH	2011	Yes	Yes	No	$100M; $500M	10; 50	None
OK	2010	Yes	Yes	No	$5M	5	125% of mean wage
OR	2015	None	None	Yes	$50M	10–35	None
PA	2022	Yes	Yes	No	$75M	25	None
SC	2012	Yes	Yes	No	$50M	None	None
TN	2016	Yes	Yes	No	$100M	15	None
TX	2013	Yes	Yes	No	$200M	20	120% of mean wage
UT	2016	Yes	Yes	No	$150M	15	110% of mean wage
VA	2009	Yes	Yes	No	$150M; $1B	50; 25	None
WA	2010	Yes	Yes	No	$25M	20	None
WI	2023	Yes	Yes	No	$150M	30	150% of mean wage
WV	2016	Yes	None	Yes	$50M	50	None
WY	2010	Yes	Yes	No	$50M	20	None

Notes: This table summarizes state-level data center incentive provisions, including the year of adoption, whether the incentive applies to state sales taxes, local sales taxes, and local property taxes, and the associated investment, employment, and wage requirements. Dollar values are reported in millions unless otherwise indicated.

Table B.3
Water adjacent vs Non-water adjacent Observable characteristics in Year 2000

Characteristic	Waterfront difference	t-statistic	p-value
Panel A: Percentage-point measures			
Minority share	0.575	1.674	0.101
Female share	0.051	0.475	0.637
Poverty rate	−0.526	−1.758	0.085
Unemployment rate	0.061	0.417	0.679
Information-sector employment share	0.053	1.635	0.109
Panel B: Logged characteristics			
Log median age	−0.006	−0.994	0.325
Log mean commute time	0.005	0.482	0.632

Notes: Each row compares waterfront and non-waterfront counties using year 2000 data. Positive differences indicate higher values in waterfront counties. Panel A is measured in percentage points. Commute time is measured in minutes.

Table B.4
Summary of State Data Center Tax Incentives

Measure	Statistic	All	Early adopters through 2016	Late adopters after 2016
Incentives adopted	Number of states	40	22	18
	Percent of adopters	100.0%	55.0%	45.0%
State sales-tax abatement	Number of states	40	22	18
	Percent of states	100.0%	100.0%	100.0%
Local sales-tax abatement	Number of states	39	21	18
	Percent of states	97.5%	95.5%	100.0%
Property-tax abatement	Number of states	8	6	2
	Percent of states	20.0%	27.3%	11.1%
Investment threshold	Number of states	36	20	16
	Percent of states	90.0%	90.9%	88.9%
	Mean threshold ($M)	$110.8	$80.9	$147.3
Employment threshold	Number of states	29	16	13
	Percent of states	72.5%	72.7%	72.2%
	Mean threshold (jobs)	17.3	13.2	22.2
Wage threshold	Number of states	8	6	2
	Percent of states	20.0%	27.3%	11.1%

Notes: Early adopters include states that adopted an incentive through 2016.

Table B.5
Data Center Growth Over Time

Year	Total DCs	New DCs	Total Space (sq. ft.)	New Space (sq. ft.)	Total Power (MW)	New Power (MW)	Median DC Space (sq. ft.)	Median Power (MW)
2000	408	22	24,697,480	24,283,881	2,885	2,847	33,000	2.195
2001	433	25	26,724,056	2,026,574	3,070	185	86,000	7.000
2002	447	14	28,046,290	1,322,235	3,261	191	48,627	4.110
2003	452	5	28,574,400	528,110	3,372	111	433,333	35.000
2004	457	5	29,419,076	844,675	3,429	57	40,000	7.000
2005	469	12	30,926,318	1,507,243	3,609	180	34,000	8.750
2006	483	14	32,226,518	1,300,200	3,759	150	85,440	10.000
2007	509	26	36,573,572	4,347,053	4,322	564	134,550	12.250
2008	550	41	41,155,872	4,582,302	4,820	498	77,300	6.000
2009	589	39	45,276,540	4,120,666	5,235	415	84,000	8.000
2010	640	51	51,163,256	5,886,719	5,883	648	64,500	6.720
2011	689	49	56,852,296	5,689,037	6,454	571	79,914	7.965
2012	752	63	64,089,392	7,237,097	7,271	818	78,760	7.000
2013	815	63	73,983,872	9,894,482	8,543	1,271	101,800	9.500
2014	882	67	82,544,432	8,560,557	9,692	1,150	80,000	10.000
2015	946	64	91,267,848	8,723,417	10,955	1,262	110,620	13.680
2016	1,036	90	105,221,632	13,953,786	13,266	2,312	147,950	24.000
2017	1,127	91	120,098,696	14,877,061	15,701	2,435	154,000	27.500
2018	1,241	114	143,640,528	23,541,834	20,214	4,513	180,000	40.000
2019	1,343	102	160,724,992	17,084,459	23,994	3,779	200,000	35.985
2020	1,442	99	176,718,304	15,993,310	28,126	4,132	200,000	42.000
2021	1,608	166	207,758,096	31,039,805	35,786	7,660	245,000	45.000
2022	1,833	225	253,243,440	45,485,332	48,582	12,796	250,000	57.760
2023	2,253	420	335,423,040	82,179,605	74,575	25,993	267,000	65.000
2024	3,413	1,160	554,545,088	219,122,047	158,137	83,562	250,750	62.505
2025	5,855	2,442	854,548,160	300,003,049	407,402	249,265	282,000	82.500

Notes: Cumulative counts, facility square footage, and power capacity are summed across counties in the geographic panel. Inventory event dates prioritize announcement, construction start, construction finish, and activation; these are not activation-only counts.

the

Table B.6
Data Center Size and Power Capacity by Inventory Event Period

Event period	*N*	Mean	Median	SD	P25	P75
Panel A: Facility size (square feet)						
Before 2016	842	108,787	58,098	148,135	23,500	138,000
2016 or later	2,831	269,933	250,000	159,869	180,000	305,200
All years	3,673	232,992	222,000	171,210	124,780	288,000
Panel B: Selected power capacity (MW)						
Before 2016	911	12.04	5.00	17.79	1.49	15.00
2016 or later	4,605	86.09	66.00	71.32	48.00	100.00
All years	5,516	73.86	60.00	71.09	34.00	90.00

Notes: The table reports facilities with selected inventory event dates between 2000 and 2025. The event date prioritizes announcement, then construction start, construction finish, and activation. The post-2016 group includes event years 2016 or later. Space is in square feet and power in MW. SD denotes standard deviation; P25 and P75 are the 25th and 75th percentiles.

Table B.7
Facility Size and Power Capacity Before and After 2016

	(1) Log facility size (square feet)	(2) Log power capacity (MW)
Inventory event in 2016 or later	1.337***	2.399***
	(0.109)	(0.166)
State fixed effects	Yes	Yes
Observations	3,668	5,514
Adjusted R^2	0.369	0.535

Notes: The sample covers selected inventory event dates from 2000 through 2025, prioritizing announcement, construction start, construction finish, and activation. The dependent variables are log facility square footage and log selected power capacity. The post-2016 indicator equals one for event years 2016 or later. Both models include state fixed effects and state-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.8
Data Center Tax Incentives, Population, and Migration

	(1)	(2)
Panel A: Through 2022	Log population	Net migration per 1,000
Post × Treated × Waterfront	-0.0023	0.2587
	(0.0024)	(0.4597)
p-value	0.353	0.576
County FE	Yes	Yes
State × Year FE	Yes	Yes
Waterfront × Year FE	Yes	Yes
Adjusted R^2	0.9995	0.3357
Observations	265,633	247,299

Notes: This table reports annual stacked triple-difference OLS estimates. Column 1 uses the natural logarithm of county population; Column 2 uses total net migration per 1,000 residents, combining domestic and international net migration. Population is measured on July 1, and migration covers the preceding July–June period. Census migration rates use the average of beginning- and end-period population as the denominator. Population observations begin in 2000; annual migration observations begin in 2001. Event year zero is the adoption year, and the window spans −5 to +5 years. Standard errors clustered by state are in parentheses. Observations count stacked county–years. Source: U.S. Census Bureau Population Estimates Program. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B.9
Additional Robustness Checks

	(1)	(2)	(3)	(4)	(5)
Panel A: Four years before and after event year zero	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Waterfront	0.2430**	0.0161*	0.0210*	0.0252**	-0.0118
	(0.0948)	(0.0095)	(0.0116)	(0.0115)	(0.0073)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.936	0.993	0.992	0.993	0.9995
Observations	210,384	225,067	225,067	225,182	762,282
Panel B: Three years before and after event year zero	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Waterfront	0.1829**	0.0119	0.0171*	0.0173*	-0.0105*
	(0.0735)	(0.0081)	(0.0090)	(0.0090)	(0.0063)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.954	0.994	0.993	0.994	0.9996
Observations	167,199	177,237	177,237	177,334	584,169
Panel C: Excluding Virginia and California	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Waterfront	0.2483**	0.0202**	0.0262**	0.0310**	-0.0108*
	(0.1232)	(0.0103)	(0.0120)	(0.0126)	(0.0065)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.923	0.992	0.991	0.992	0.9996
Observations	248,353	269,912	269,912	270,047	580,368

Table B.9 (continued)

Panel D: Standard errors clustered by state and year	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Waterfront	0.3032**	0.0203**	0.0248**	0.0308***	-0.0105*
	(0.1252)	(0.0094)	(0.0115)	(0.0119)	(0.0063)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.919	0.992	0.991	0.992	0.9996
Observations	251,279	271,501	271,501	271,637	584,169
Panel E: Excluding the 10 largest counties	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Waterfront	0.2468**	0.0185*	0.0228*	0.0286**	-0.0047
	(0.1196)	(0.0102)	(0.0121)	(0.0117)	(0.0037)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.909	0.992	0.991	0.992	0.9995
Observations	250,751	270,877	270,877	271,013	582,744
Panel F: Excluding the 20 largest counties	Normalized data center count	Total revenue	Total taxes	Total expenditure	Employment
Post × Treated × Waterfront	0.2706**	0.0124	0.0161	0.0261**	-0.0035
	(0.1156)	(0.0122)	(0.0123)	(0.0121)	(0.0034)
County FE	Yes	Yes	Yes	Yes	Yes
State × Time FE	Yes	Yes	Yes	Yes	Yes
Waterfront × Time FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2 / Pseudo R^2	0.900	0.991	0.990	0.991	0.9994
Observations	249,983	270,007	270,007	270,143	580,787

Notes: Each panel reports separate stacked triple-difference regressions. Column 1 uses annual OLS through 2024; the data center count is divided by its 1996–2024 county–year mean (0.26691), and event year zero is the year of adoption. Columns 2–4 use annual PPML through 2022, with adoption as event year zero. Columns 2–3 require total revenue, total taxes, general charges, and intergovernmental revenue to be reported. Column 5 uses quarterly PPML for average private employment. Time fixed effects refer to years in Columns 1–4 and quarters in Column 5; all fixed effects are interacted with cohort. All robustness samples exclude California. Panels A–B use event windows of four and three years on either side of each model's treatment reference. Panel C also excludes Virginia in every column; Texas is retained. Standard errors are clustered by state except in Panel D, which clusters by state and year. Panels E–F retain each model's original county exclusion lists; employment exclusions use the July 1, 2025 population ranking. Parentheses report standard errors. Fit statistics are adjusted R^2 for OLS and pseudo R^2 for PPML. Observations count stacked county–years in Columns 1–4 and county–quarters in Column 5. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

C Where Data Centers Locate

Data center location decisions depend on both the type of facility and the services it provides. All data centers require electricity, cooling, land, and digital connectivity, but not every facility must remain close to its users. Location choices therefore reflect trade-offs among operating costs, proximity to customers, performance requirements, regulatory considerations, and infrastructure availability.

C.0.1 Proximity to Users

The importance of proximity varies across data center types. Proximity is particularly important for latency, access, and data security. Each additional 100 kilometers between a user and a server adds approximately 1 millisecond to data transmission time. For some customers, such as hospitals and financial institutions, even a few milliseconds can be valuable. Therefore, they may rely more on urban facilities. In contrast, hyperscale and cloud providers have greater geographic flexibility because they provide computing services remotely. These facilities primarily cater to tasks such as batch processing, data storage, and AI model training, which are less sensitive to small transmission delays.

C.0.2 Electricity Availability and Cost

Abundant and reliable electricity is an important input for data centers, and location decisions are often based on the availability of inexpensive electricity. They not only consume large amounts of electricity to run their servers but also to cool the data center. Data centers operate around the clock, and even brief disruptions can be costly. The local grid must have sufficient generation and transmission capacity to accommodate their substantial and persistent electricity demand.

To attract data center investments, some localities have invested in new power plants and transmission lines to ensure sufficient electricity supply. In other cases, utilities have

expanded existing infrastructure or planned new investments in anticipation of growing data center demand.

C.0.3 Water and Cooling Resources

The Servers in a data center produce a considerable amount of heat, which must be removed continuously for the data center to operate safely. There are several cooling technologies that data centers can use. Servers produce considerable heat and must be cooled continuously to operate safely. Although data centers use several cooling technologies, many facilities rely partly on water-based systems because of their efficiency. Access to dependable water supplies may therefore make locations near rivers, lakes, reservoirs, and other water sources particularly attractive.

At the same time, data center development can place additional pressure on local water systems, especially during droughts or periods of high residential demand. Communities hosting large facilities may need to expand water-treatment, distribution, or wastewater infrastructure to accommodate this additional demand.

C.0.4 Digital Connectivity

Data centers require high-capacity internet connections and access to fiber-optic networks to transmit large volumes of information quickly and reliably. Locations near major fiber routes and internet exchange points offer lower latency and greater network redundancy.

Connectivity is particularly important for colocation and edge facilities serving hospitals, financial firms, and other customers requiring immediate access to data. Although large cloud facilities can operate farther from their users, they still require extensive fiber infrastructure to provide reliable computing and storage services.

C.0.5 Land and Physical Infrastructure

Large cloud and hyperscale data centers require substantial parcels of land for server buildings, electrical substations, cooling systems, backup generators, and security infrastructure. These facilities often locate outside dense urban areas, where land is less expensive and large industrial sites are more readily available.

Proximity to transmission lines, electrical substations, fiber networks, highways, and other supporting infrastructure can reduce construction costs and development delays. Local zoning rules and the speed of the permitting process can also affect whether a proposed location is feasible.

C.0.6 Tax Incentives and Government Support

State and local governments frequently offer property-tax abatements, sales-tax exemptions, infrastructure assistance, and expedited permitting to attract data center investment. These incentives can influence the selection among locations with otherwise similar characteristics.

However, financial incentives cannot compensate for inadequate physical infrastructure. For policymakers seeking to attract data centers, access to reliable electricity, sufficient water resources, and high-capacity internet generally matters more than tax incentives. Moreover, generous incentives may reduce the direct fiscal benefits that communities receive from hosting these facilities, even when local governments must finance additional supporting infrastructure.

C.0.7 Climate and Environmental Risk

Climate conditions affect both cooling costs and operational reliability. Cooler climates can reduce the energy required to maintain safe server temperatures. Firms also consider exposure to floods, hurricanes, wildfires, droughts, and extreme heat because interruptions to data center operations can be extremely costly.

Facilities therefore require backup power, redundant network connections, and emergency plans, particularly when located near bodies of water or in regions exposed to natural hazards. These environmental considerations may create a tradeoff: access to water can improve cooling efficiency but may also increase exposure to flooding or competition over scarce water supplies.

Overall, data centers are not distributed randomly across locations. Urban colocation and edge facilities tend to remain close to customers with latency and compliance requirements, whereas large cloud, hyperscale, and artificial-intelligence facilities have greater flexibility to locate in lower-cost areas. Across all types, reliable electricity, water availability, high-capacity internet, and supporting infrastructure remain central to location decisions. These requirements also imply that the local consequences of data center development may be concentrated in communities that must expand or upgrade their electricity, water, and communications infrastructure.

D Types of Data Centers

Data centers differ in their ownership, scale, location, and the computing services they provide. These differences shape their infrastructure requirements, proximity to users, and effects on local electricity and water systems. Understanding the different types of data centers therefore helps explain why some facilities locate near major urban markets, while others locate in lower-cost areas with abundant land and natural resources.

D.1 Enterprise Data Centers

Enterprise data centers are owned and operated by individual organizations for their internal computing needs. Large corporations, financial institutions, hospitals, universities, and government agencies may maintain these facilities to exercise direct control over their data, security, and regulatory compliance. Enterprise data centers are often located on the organization's property or near its principal operations. However, many organizations have moved part of their computing activity to colocation centers or cloud providers.

D.2 Third-Party Colocation Centers

Third-party colocation centers provide physical space, electricity, cooling, connectivity, and security to multiple customers. Rather than constructing an independent facility, a firm rents space and installs its own servers and networking equipment inside a shared data center. The customer generally retains control over its equipment, while the colocation provider manages the building and its supporting infrastructure.

Colocation centers are frequently located near the firms they serve. Hospitals, financial institutions, and other organizations with strict latency, security, or regulatory requirements may prefer urban colocation facilities that provide rapid and dependable access to their computing systems. Because these facilities serve multiple customers, they can also provide connections to several telecommunications networks and cloud platforms from a

single location.

D.3 Cloud Data Centers

Cloud data centers are operated by providers that sell computing power, storage, databases, and software as services over the internet. Customers purchase these resources when needed rather than owning and maintaining the underlying physical equipment. Because users access cloud services remotely, cloud facilities do not always need to be located near individual customers.

This flexibility allows cloud providers to construct large facilities in areas with inexpensive land, abundant electricity, sufficient water, and strong fiber-optic connectivity. Batch-processing operations, data storage, and other activities that are relatively insensitive to small transmission delays can be performed efficiently from these lower-cost hubs.

D.4 Hyperscale Data Centers

Hyperscale data centers are exceptionally large facilities designed to support the operations of major cloud-computing and technology companies. They contain large numbers of servers and are designed to expand computing capacity rapidly as demand increases. Although many hyperscale facilities provide cloud services, the term "hyperscale" refers primarily to their size and scalability rather than to a distinct service model.

These facilities require substantial electrical capacity, large parcels of land, extensive cooling infrastructure, and high-capacity network connections. Consequently, they can have especially pronounced effects on local electricity grids, water systems, land use, and infrastructure investment.

D.5 Edge Data Centers

Edge data centers are smaller facilities located close to users or connected devices. Their purpose is to process information near the point where it is generated rather than transmit-

ting everything to a distant cloud facility. This reduces latency and supports applications requiring nearly immediate responses, including telecommunications services, autonomous systems, online gaming, and certain healthcare and financial activities.

Although an individual edge facility is smaller than a cloud or hyperscale center, a provider may operate many such facilities across different geographic markets. Their location decisions therefore emphasize proximity to users and network connectivity more heavily than access to large quantities of inexpensive land.

D.6 Artificial-Intelligence and High-Performance Computing Facilities

Artificial-intelligence and high-performance computing facilities support computationally intensive workloads, including model training, scientific simulations, and large-scale data analysis. These workloads often use specialized processors and require substantially more electricity per server than traditional computing activities. Their high equipment density also produces considerable heat, increasing the need for advanced cooling systems.

Some artificial-intelligence applications require quick responses and therefore benefit from being located near users. However, artificial-intelligence model training and many batch-processing workloads are less sensitive to distance. These activities can operate more efficiently from large, lower-cost cloud hubs with abundant power and cooling resources.

The boundaries among these categories are not always sharp. A hyperscale facility may provide cloud services and artificial-intelligence computing, while a colocation center may lease space to cloud providers as well as hospitals and financial firms. Nevertheless, these distinctions help explain the different considerations involved in data center location decisions.

E Theory: Dynamic Local Effects of Data Center Expansion

This section develops a dynamic model of how large data center projects affect local employment, resource use, and welfare. The model studies when subsidies and demand driven expansions create net local job gains rather than reallocating employment across sectors.

The model features a scalable data center, non-data center industries that differ in resource intensity and complementarity, households that supply labor and bear local costs, and a government choosing subsidies under demand uncertainty. The main result is that employment gains are not guaranteed: when demand is high and local resources are constrained, data center growth can crowd out other industries and reduce welfare even when the project appeared beneficial ex ante.

E.1 Setup

Consider a locality populated by heterogeneous households, a data center sector indexed by D, and a collection of non-data center industries indexed by $j \in \mathcal{J}$. Time is discrete, $t = 0, 1, 2, \ldots$. The aggregate local state is given by the demand state

$$\theta_t,$$

where θ_t captures the strength of demand for data center activity (for example, AI compute demand or major shifts in the demand for cloud and related digital infrastructure).

The demand state evolves according to

$$\theta_{t+1} \sim \mathcal{P}(\cdot \mid \theta_t), \tag{E.1}$$

where $\mathcal{P}$ is a Markov transition kernel that captures how demand for data center services

changes over time.

Households can supply up to $\bar{N}_t$ units of labor in each period. Let N_t^D denote data center employment and N_t^O denote total employment in all other local industries. Total employment and unemployment are

$$N_t = N_t^D + N_t^O, \qquad U_t = \bar{N}_t - N_t \geq 0. \tag{E.2}$$

The labor supply ceiling $\bar{N}_t$ is exogenous in the short run.

E.2 Data Center Production, Profits, Subsidies, and Adjustment

The data center chooses labor n_{Dt}, resource use r_{Dt}, and investment i_{Dt}. Capital evolves according to

$$k_{D,t+1} = (1 - \delta_D)k_{Dt} + i_{Dt}, \tag{E.3}$$

where $\delta_D \in (0, 1)$ is the depreciation rate.

Data center output is determined by a standard production function

$$y_{Dt} = \tilde{F}_D(k_{Dt}, n_{Dt}, r_{Dt}), \tag{E.4}$$

where $\tilde{F}_D$ maps capital, labor, and local resource use into output. The physical production technology does not depend on the demand state.

Instead, the demand state enters on the revenue side. Let the output price of data center services be

$$p_t^D = p^D(\theta_t), \tag{E.5}$$

where $p^D(\theta_t)$ is increasing in θ_t. A higher demand state therefore raises the revenue generated by a given level of output and, through that channel, increases the incentive to invest and expand.

Local policy includes a project-specific subsidy with generosity $\sigma_t \in \mathbb{R}_+$, where larger

σ_t denotes a more generous package of tax abatements, cash grants, and infrastructure support. The subsidy enters the data center's payoff via a transfer function $S_D(\sigma_t, k_{Dt}, i_{Dt})$.

Period t profits are

$$\pi_{Dt} = p^D(\theta_t)\, y_{Dt} - w_t n_{Dt} - q_t r_{Dt} - \rho_t i_{Dt} - \Gamma_D(i_{Dt}, k_{Dt}) + S_D(\sigma_t, k_{Dt}, i_{Dt}), \tag{E.6}$$

where w_t is the local wage, q_t is the local resource price, ρ_t is the purchase price of installed capital, $\Gamma_D(i, k)$ is an adjustment cost, and $S_D(\sigma_t, k_{Dt}, i_{Dt})$ is the subsidy or policy transfer.

Assumption 1 (Data center technology, demand, generosity, and adjustment).

1. $\tilde{F}_D$ is increasing and concave in capital k_{Dt}, labor n_{Dt}, and resources r_{Dt}.
2. $p^D(\theta)$ is increasing in θ.
3. $\Gamma_D(i, k)$ is nonnegative, convex in i, continuously differentiable, and satisfies $\Gamma_D(0, k) = 0$.
4. $S_D(\sigma_t, k, i)$ is increasing in policy generosity σ_t, so that more generous subsidies raise the effective return to installed capital and investment.

Convex adjustment costs imply that changing scale rapidly is costly, which delivers gradual rather than instantaneous expansion paths. Because the demand state raises the effective output price $p^D(\theta_t)$, higher θ_t increases the marginal value of expanding data center capacity. Throughout the model, θ_t affects local outcomes through the value of data center output and the induced expansion path for k_{Dt}, not through the underlying physical production technology.

E.3 Industry Heterogeneity and Resource Use

Each non-data center industry $j \in \mathcal{J}$ is characterized by two reduced form parameters:

- resource intensity $\alpha_j^R \geq 0$, which measures how strongly the industry's employment is exposed to local resource prices;

- complementarity $\alpha_j^C \geq 0$, which measures how strongly the industry's employment benefits from the presence or scale of data center activity.

Industry j uses capital K_{jt}, labor N_{jt}, and local resources R_{jt} to produce

$$Y_{jt} = F_j(K_{jt}, N_{jt}, R_{jt}), \tag{E.7}$$

where F_j is increasing and concave in its inputs.

Industry j solves

$$\max_{K_{jt},\, N_{jt},\, R_{jt}} \left\{ p_{jt} Y_{jt} - r_t K_{jt} - w_t N_{jt} - q_t R_{jt} \right\}, \tag{E.8}$$

which yields labor and resource demand schedules

$$N_{jt} = N_j(w_t, r_t, q_t, k_{Dt}; \alpha_j^R, \alpha_j^C), \qquad R_{jt} = R_j(w_t, r_t, q_t; \alpha_j^R, \alpha_j^C).$$

Dependence on k_{Dt} in labor demand is reduced form and captures demand and productivity linkages whereby larger data center scale raises activity in complementary industries, beyond the indirect effect operating through local resource prices.

Aggregate employment and resource use outside the data center sector are

$$N_t^O = \sum_{j \in \mathcal{J}} N_{jt}, \qquad R_t^O = \sum_{j \in \mathcal{J}} R_{jt}. \tag{E.9}$$

Let R_{ht} denote household h's use of local resources (electricity, water, land, and related services), and define aggregate household resource use as

$$R_t^H = \sum_{h \in \mathcal{H}} R_{ht}. \tag{E.10}$$

Total local resource use is

$$R_t = r_{Dt} + R_t^O + R_t^H = r_{Dt} + \sum_{j \in \mathcal{J}} R_{jt} + \sum_{h \in \mathcal{H}} R_{ht}. \tag{E.11}$$

We collapse scarcity, congestion, and stress into a single reduced form harm function by letting aggregate resource use R_t generate a scalar stress cost:

$$\Omega_t = \Omega(R_t), \tag{E.12}$$

where $\Omega(R_t)$ summarizes congestion, reliability pressure, environmental stress, and unpriced local harms from heavy resource use.

Assumption 2 (Industry heterogeneity and spillovers).

1. $\Omega(R)$ is increasing and convex in R.
2. $\dfrac{\partial N_j}{\partial q_t} < 0$ for all j, and $\left|\dfrac{\partial N_j}{\partial q_t}\right|$ is increasing in α_j^R.
3. $\dfrac{\partial N_j}{\partial k_{Dt}} \geq 0$ for all j, and $\dfrac{\partial N_j}{\partial k_{Dt}}$ is increasing in α_j^C.

Assumption 2 implies that all industries are weakly harmed by higher local resource prices, but highly resource-intensive industries are harmed more. At the same time, industries with stronger complementarities to data center scale benefit more from expansion in k_{Dt}.

E.4 Labor Market Closure

To permit unemployment in the simplest possible way, assume a downward rigid wage floor $w_t \geq \bar{w}_t$. Employment is demand determined:

$$N_t^D = \mathcal{N}^D(w_t, q_t, \theta_t, k_{Dt}), \qquad N_t^O = \sum_{j \in \mathcal{J}} N_j(w_t, r_t, q_t, k_{Dt}; \alpha_j^R, \alpha_j^C). \tag{E.13}$$

Total employment and unemployment are given by equation (E.2). In this closure, data center expansion can increase direct employment while simultaneously reducing employment in some industries through higher local factor prices and increasing employment in others through complementary demand or production linkages. Because wages cannot fall below $\bar{w}_t$, weak labor demand shows up as unemployment rather than lower wages.

E.5 Households

Households are indexed by $h \in \mathcal{H}$ and are heterogeneous in labor endowments, asset holdings, exposure to local prices, and housing positions. Household h chooses consumption c_{ht}, labor supply n_{ht}, and savings $b_{h,t+1}$ to maximize expected discounted utility

$$U_h = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u_h(c_{ht}, 1 - n_{ht}, a_{ht}, \Omega(R_t)) \right], \tag{E.14}$$

where c_{ht} is consumption, $1 - n_{ht}$ is leisure, a_{ht} captures amenities or local quality of life, $\Omega(R_t)$ captures local stress from resource use, and $\beta \in (0, 1)$ is the discount factor.

The household faces a period budget constraint

$$c_{ht} + b_{h,t+1} = w_t n_{ht} + (1 + r_t) b_{ht}, \tag{E.15}$$

where w_t is the local wage, r_t is the local interest rate on savings, and b_{ht} is beginning-of-period financial assets.

Assumption 3 (Household heterogeneity and welfare exposure).

1. Period utility u_h is increasing in consumption and amenities and decreasing in labor effort and stress $\Omega(R_t)$.

2. Households differ in wages, asset holdings, and exposure to stress (through $\Omega(R_t)$).

3. Household policy functions (choices of c_{ht}, n_{ht}, and $b_{h,t+1}$) are continuous in prices.

Households internalize the impact of resource use R_t through both utility and prices. Resource-intensive expansion affects utility directly via $u_h(\cdot, \Omega(R_t))$ and indirectly by shifting w_t, q_t, and possibly a_{ht}.

E.6 Welfare and the Government Problem

Per-period local welfare is defined as

$$\mathcal{W}(\theta_t, \sigma_t) = \sum_{h \in \mathcal{H}} \omega_h \, u_h(c_{ht}, 1 - n_{ht}, a_{ht}, \Omega(R_t)) + \sum_{j \in \mathcal{J}} \omega_j \, Y_{jt} + \omega_D y_{Dt} - \Omega(R_t), \tag{E.16}$$

where:

- ω_h are household welfare weights,
- Y_{jt} is output in non-data center industry j,
- y_{Dt} is output of the data center,
- ω_j and ω_D are weights on industry and data center production outcomes,
- $\Omega(R_t)$ is a reduced form stress cost capturing congestion, reliability pressure, environmental stress, and unpriced local harms from heavy resource use.

The government chooses subsidy generosity σ_t to solve

$$W(\theta_t) = \max_{\sigma_t} \left\{ \mathcal{W}(\theta_t, \sigma_t) - C^{\text{fisc}}(\sigma_t) + \beta \, \mathbb{E}\left[W(\theta_{t+1}) \mid \theta_t, \sigma_t \right] \right\}, \tag{E.17}$$

where $C^{\text{fisc}}(\sigma_t)$ denotes the fiscal cost of subsidies, tax expenditures, and infrastructure spending associated with generosity σ_t. Since the demand state affects welfare through the price of data center output and the induced expansion of capacity, the planner's subsidy rule is shaped by expectations over future demand realizations rather than by any direct demand effect on physical technology.

Assumption 4 (Welfare and public finance).

1. $\Omega(R)$ is increasing and convex in resource use.

2. Fiscal costs $C^{\text{fisc}}(\sigma)$ are increasing and convex in subsidy generosity.

3. The planner attaches positive weight to household welfare and to local production outcomes, so that all weights $\{\omega_h\}_{h\in\mathcal{H}}$, $\{\omega_j\}_{j\in\mathcal{J}}$, and ω_D are nonnegative.

Under these assumptions, the government chooses a subsidy rule $\{\sigma_t\}$ based on the ex ante distribution of future demand states $\{\theta_t\}$ and the induced distribution of household welfare. Realized welfare is state contingent and can differ sharply from ex ante expectations in high demand, high scaling states, especially when resource-intensive industries are large and local stress costs are steep.

E.7 Employment Responses and Crowding Out

The key employment margin in the model is the response of total employment to a subsidy induced or demand induced expansion in data center activity. Let

$$\Delta N_t = \Delta N_t^D + \sum_{j\in\mathcal{J}} \Delta N_{jt}, \tag{E.18}$$

where ΔN_t^D is the direct change in data center employment and ΔN_{jt} is the change in employment in industry j.

For each industry j, a first order approximation implies

$$\Delta N_{jt} \approx \frac{\partial N_j}{\partial k_{Dt}} \Delta k_{Dt} + \frac{\partial N_j}{\partial q_t} \Delta q_t. \tag{E.19}$$

The first term captures gains from complementarity with data center scale, while the second captures losses from higher local resource prices.

Lemma 1 (Gradual expansion)**.** *Under Assumption 1, an increase in subsidy generosity σ_t or in the demand state θ_t weakly raises the shadow value of installed data center capital and therefore weakly raises optimal investment i_{Dt}.*

Sketch. A more generous subsidy raises the payoff associated with installed capital through $S_D(\sigma_t, k_{Dt}, i_{Dt})$. A higher demand state raises the output price $p^D(\theta_t)$ and therefore the marginal revenue from a given level of output. By concavity of production and convexity of Γ_D, the first order condition for investment equates the marginal value of additional capital to the marginal purchase and adjustment cost. Since marginal adjustment cost is increasing in i_{Dt}, positive shocks to σ_t or θ_t increase investment but do not induce an infinite jump in k_{Dt}. Hence expansion is monotone and gradual. □

Lemma 2 (Stress transmission)**.** *If data center investment raises k_{Dt} and resource use r_{Dt}, then the stress cost Ω_t weakly rises.*

Sketch. By equation (E.11), an increase in data center scale raises total resource use R_t through higher r_{Dt}. Since equation (E.12) implies that Ω is increasing in R_t, higher R_t weakly raises Ω_t. □

Proposition 1 (Industry-level employment response)**.** *Let industry j be characterized by resource intensity α_j^R and complementarity α_j^C. Following a subsidy or demand shock that raises data center scale ($\Delta k_{Dt} > 0$) and local resource prices ($\Delta q_t > 0$), industry j employment satisfies equation* (E.19)*. Then:*

1. *$\Delta N_{jt} > 0$ is more likely when complementarity α_j^C is large, resource intensity α_j^R is small, and Δq_t is modest.*

2. *$\Delta N_{jt} \leq 0$ is more likely when resource intensity α_j^R is large, complementarity α_j^C is small, and Δq_t is large.*

Sketch. By Assumption 2, the positive term $\frac{\partial N_j}{\partial k_{Dt}} \Delta k_{Dt}$ is larger for industries with higher complementarity α_j^C, while the negative term $\frac{\partial N_j}{\partial q_t} \Delta q_t$ is more negative for industries with higher resource intensity α_j^R. The sign of ΔN_{jt} therefore depends on whether the complementary gain from higher data center scale dominates the price induced employment loss. □

Proposition 2 (Net employment gains versus crowding out)**.** *Suppose a subsidy or demand shock raises data center activity so that* $\Delta N_t^D > 0$ *and induces changes in non-data center industry employment according to equation* (E.19)*. Then:*

1. *Net local employment grows if and only if*

$$\Delta N_t^D + \sum_{j \in \mathcal{J}} \Delta N_{jt} > 0.$$

2. *Net local employment does not grow if and only if*

$$\Delta N_t^D + \sum_{j \in \mathcal{J}} \Delta N_{jt} \leq 0.$$

Equivalently, crowding out dominates whenever

$$\left| \sum_{j \in \mathcal{J}} \Delta N_{jt} \right| \geq \Delta N_t^D \qquad \textit{when} \qquad \sum_{j \in \mathcal{J}} \Delta N_{jt} < 0. \tag{E.20}$$

Sketch. Equation (E.18) gives the decomposition of aggregate employment into direct data center employment and the sum of industry-level employment changes outside the data center sector. Net employment rises if the direct employment gain and the employment gains in complementary industries jointly exceed the losses in resource-intensive industries. It does not rise when these losses weakly dominate. □

Proposition 3 (Conditions for net employment growth)**.** *Holding fixed the path of demand shocks* $\{\theta_t\}$*, net local employment is more likely to grow after a subsidy or favorable demand shift when:*

1. *data center production is relatively labor intensive, so that* $\dfrac{\partial N_t^D}{\partial \theta_t}$ *and* $\dfrac{\partial N_t^D}{\partial \sigma_t}$ *are large;*
2. *local resource supply is relatively elastic, so that* $\dfrac{\partial q_t}{\partial R_t}$ *is small;*

3. *the local industry composition places greater weight on industries with high complementarity α_j^C and lower weight on industries with high resource intensity α_j^R;*

4. *the local labor market begins with slack, so that wage pressures from expansion are limited.*

Conversely, net employment is less likely to grow when data centers are highly capital and resource intensive, local resource supply is inelastic, and the local economy contains many industries that are strongly exposed to local resource prices but weakly complementary to data center activity.

Sketch. The direct employment gain ΔN_t^D is larger when data centers are more labor intensive and when subsidy generosity or stronger demand translates strongly into scale. The aggregate indirect employment response $\sum_j \Delta N_{jt}$ is less negative, or more positive, when the price response Δq_t to resource use is muted and when a larger share of industries has high α_j^C rather than high α_j^R. Slack labor markets additionally limit wage driven crowding out in non-data center industries. □

Proposition 4 (Ex ante support and ex post regret)**.** *Suppose policymakers evaluate a project using expected welfare before the realization of θ_t, while realized welfare is state contingent. If low and moderate realizations of θ_t generate manageable stress costs but high realizations generate sharply increasing stress through the convex function $\Omega(R)$, then there exist parameter values such that expected ex ante welfare from subsidizing the project is positive while ex post welfare is negative in high demand states.*

Sketch. Consider the period welfare expression in equation (E.16). Benefits from expansion rise with the demand induced value of data center output, direct employment, and employment gains in complementary industries, but local costs enter through $\Omega(R_t)$, which is convex in R_t. Convexity implies that as the high demand state pushes resource use far enough upward, marginal and total stress costs can increase more than proportionally. One

can choose a distribution for θ_t such that expected welfare, averaging across states, remains positive, yet realized welfare in sufficiently high states becomes negative. □

E.8 Testable Implications

The model yields a set of empirical predictions that map naturally to industry level heterogeneity.

- **Direct versus net jobs.** Announcements, subsidies, or openings should increase employment within the data center sector, but total local employment rises only if the direct gain plus the gains in complementary industries exceed the losses in resource-intensive industries.
- **Resource intensity heterogeneity.** Industries with higher measured resource intensity should experience weaker employment growth, or outright employment declines, after data center entry or subsidy adoption.
- **Complementarity heterogeneity.** Industries with higher measured complementarity to data centers should experience stronger employment growth after data center expansion.
- **Industry mix as a moderator.** Places with an industry mix tilted toward high α_j^R industries should exhibit more crowding out and smaller net employment gains, while places tilted toward high α_j^C industries should exhibit stronger net employment growth.
- **Demand state heterogeneity.** Locations exposed to larger demand driven expansions in data center scale should experience larger direct employment gains but also stronger pressure on local resources and therefore greater scope for crowding out.
- **Stronger effects where resource supply is less elastic.** Net employment, wage, and price effects should be larger in places where local resource prices respond more sharply to increases in aggregate resource use.